

# Sri Lanka Institute of Information Technology



IT3101 – Data Warehousing and Business Intelligence  
Year 3, Semester 1- 2026

## Final Report

| Group ID 2026-DS-52 |                       |
|---------------------|-----------------------|
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## Member 1 Deliverable: Tasks 1 and 2

*Prepared by: Rifshadh M.R.M. (Member 1)*

### Task 1: Dataset Selection and Business Scenario Identification

#### Dataset Overview

**Business domain:** Banking and Finance.

**Purpose of the system:** The dataset represents the core operational database of a retail bank, covering client accounts, account holders, standing payment orders, individual transactions, loans and credit cards, together with demographic data for the districts the bank operates in. In production, this kind of system supports the bank's day to day operations: opening accounts, processing transactions, issuing loans and cards, and tracking account activity.

**Business problem addressed:** The bank's operational database is optimised for recording individual transactions, not for analysis. Management has no consolidated view of account activity trends, loan performance, or regional differences in customer behaviour. This makes it difficult to answer questions such as which districts generate the most deposits and withdrawals, how account balances trend over time, whether loan repayment status relates to account activity, and which customer segments hold cards versus loans. A data warehouse and BI layer built on top of this operational data addresses that gap by consolidating the data into a form suited to trend analysis, segmentation and executive reporting.

**Dataset source:** The PKDD'99 Financial Dataset, also known as the Berka dataset, originally prepared by Petr Berka and Marta Sochorova for the PKDD'99 Discovery Challenge, using real anonymised data from a Czech bank. The team obtained the data from the Kaggle mirror "1999 Czech Financial Dataset" (uploaded by mariammariamr), downloaded as CSV files covering all eight tables. To satisfy the assignment's multiple source type requirement, the team then restructured card into JSON and loan into Excel, and separated district out as its own source, while the remaining five tables stayed grouped as CSV. A fifth, genuinely external source, a public holidays API, was added separately to enrich the Date dimension. The exact restructuring and the reasoning behind it is set out in full in Task 2.

**Dataset format:** Five sources in total, spanning four file formats: CSV (used for two separate sources, the core tables and the district reference file), a Microsoft Excel workbook, a JSON file, and a REST API, together covering the same eight related banking tables plus the holiday data: account, client, disp (disposition), order (permanent order), trans (transaction), loan, card and district.

## Number of records

| Table    | Format (Source)  | Record count |
|----------|------------------|--------------|
| account  | CSV (Source 1)   | 4,500        |
| client   | CSV (Source 1)   | 5,369        |
| disp     | CSV (Source 1)   | 5,369        |
| district | CSV (Source 4)   | 77           |
| order    | CSV (Source 1)   | 6,471        |
| trans    | CSV (Source 1)   | 1,056,320    |
| loan     | Excel (Source 2) | 682          |
| card     | JSON (Source 3)  | 892          |

## Attributes description

| Table    | Key attributes                                                                                                         |
|----------|------------------------------------------------------------------------------------------------------------------------|
| account  | account_id, district_id, frequency of statement issuance, account creation date                                        |
| client   | client_id, birth date, gender, district_id                                                                             |
| disp     | disp_id, client_id, account_id, type (owner or disponent)                                                              |
| order    | order_id, account_id, recipient bank and account, amount, payment purpose                                              |
| trans    | trans_id, account_id, date, type, operation, amount, balance after transaction, characterisation                       |
| loan     | loan_id, account_id, date granted, amount, duration, monthly payment, status                                           |
| card     | card_id, disp_id, type (classic, junior, gold), issue date                                                             |
| district | district_id, name, region, population, number of municipalities, unemployment rate, average salary, entrepreneur ratio |

## Why This Dataset Suits a Data Warehouse and BI Solution

The dataset represents a genuine OLTP system rather than a pre aggregated or synthetic one, since it captures individual banking transactions as they occurred. It has sufficient volume for meaningful analysis, with over a million transaction records against a few thousand accounts and clients. It also offers a wide range of attributes across several natural dimensions: time (transaction and loan dates), geography (district), customer demographics (client age and gender), and product type (card type, loan status), which map cleanly onto Date, Location, Customer and Product style dimension tables. The trans table is a natural candidate for the central fact table, with loan as a possible second fact table for a loan performance data mart. Finally, the business questions this data can answer, such as regional deposit patterns, account activity trends and loan risk by segment, are realistic banking analytics concerns, which supports a credible business scenario for the rest of the assignment.

## Task 2: Data Source Identification and Preparation

The team is using five sources feeding into the same warehouse, spanning four file formats. Sources 1 to 4 all originate from the same single Kaggle download of the Berka dataset, restructured by the team into different formats and treated as separate sources. Source 5 is a wholly separate, genuinely external API. This combination satisfies the assignment's multiple source type requirement while keeping the data's real origin transparent, as set out below.

### Note on Data Origin

*Important: Sources 1 to 4 all trace back to the same Kaggle download. The team restructured the card and loan tables into JSON and Excel, and split district out as its own source, purely to create genuine source diversity for this task; the underlying values are identical to the original CSV export throughout.*

### Source 1: CSV Files

**Description:** The core set of operational tables, downloaded from the Kaggle mirror of the Berka dataset and kept in their original CSV format: account, client, disp, order and trans. This is the largest source by both table count and row count, and represents the bulk of the bank's day to day operational data.

**Source format:** Flat CSV files, comma delimited, one file per table.

### Available attributes

| Table   | Available attributes                                                                                                                                      |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| account | account_id, district_id, statement frequency, account creation date                                                                                       |
| client  | client_id, birth date, gender, district_id                                                                                                                |
| disp    | disp_id, client_id, account_id, disposition type (owner or disponent)                                                                                     |
| order   | order_id, account_id, recipient bank code, recipient account, amount, payment purpose                                                                     |
| trans   | trans_id, account_id, date, transaction type, operation, amount, balance after transaction, characterisation, bank and account of the transaction partner |

### Source 2: Excel File (loan.xlsx)

**Description:** The loan table, restructured by the team from the original Kaggle CSV export into a Microsoft Excel workbook. This represents a plausible real world scenario in which a credit or loans department maintains its own approval and status records in a spreadsheet rather than directly in the core banking database.

**Source format:** Microsoft Excel workbook (.xlsx), a single sheet with one row per loan.

### Available attributes

| Table | Available attributes                                                            |
|-------|---------------------------------------------------------------------------------|
| loan  | loan_id, account_id, loan grant date, amount, duration, monthly payment, status |

### Source 3: JSON File (card.json)

**Description:** The card table, restructured by the team from the original Kaggle CSV export into a JSON file. This represents a plausible real world scenario in which card issuance is handled by a separate card processing vendor that supplies data as a JSON feed rather than granting direct access to the core banking database.

**Source format:** JSON file, an array of card objects, one object per card.

### Available attributes

| Table | Available attributes                                            |
|-------|-----------------------------------------------------------------|
| card  | card_id, disp_id, card type (classic, junior, gold), issue date |

### Source 4: District Reference File (district.csv)

**Description:** Demographic and regional data for each district, kept as its own source even though it is also delivered as CSV. Static reference data of this kind is realistically maintained by a national statistics office or a bank's demographics team, separately from the account, client and transaction records in Source 1, which is why the team treats it as a distinct source rather than folding it into Source 1.

**Source format:** Flat CSV file, comma delimited. Confirm the character encoding on your actual download, since the Czech district names use diacritics that some tools misread.

### Available attributes

| Table    | Available attributes                                                                                                                                                                                                                                                                         |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| district | district_id, district name, region, number of inhabitants, number of municipalities by size band, number of cities, ratio of urban inhabitants, average salary, unemployment rate (1995 and 1996), number of entrepreneurs per 1,000 inhabitants, number of crimes committed (1995 and 1996) |

### Relationship Between Sources

Even though Sources 1 to 4 differ in format or in how the team has chosen to treat them, they share the exact same relational structure as the original Kaggle export, since restructuring only changed the container, not the values or the keys inside it. Within that shared structure, the tables connect as follows:

- client to disp to account: a many to many relationship, since an account can have multiple disponents and a client could in principle be linked to more than one account. All three tables are in Source 1 (CSV).
- account to order and trans: one to many, since one account has many orders and transactions. Both are in Source 1 (CSV).
- account to loan: one to many, at most one active loan per account. loan is sourced from Source 2 (Excel) but still joins to account via account\_id exactly as it would if it were still a CSV file.
- disp to card: one to one for accounts that have issued a card. card is sourced from Source 3 (JSON) but still joins to disp via disp\_id exactly as it would if it were still a CSV file.
- account and client to district: both link via district\_id. district is sourced from Source 4, a separate CSV source to Source 1, even though both happen to be CSV files, since district represents externally sourced demographic reference data rather than the bank's own operational records. This is the clearest example in the project of two sources sharing a file format while still being genuinely distinct in origin and purpose.

## Data Preparation Steps

The following preparation was carried out on the raw downloads before handing the files to Member 3 for the ETL pipeline.

1. Downloaded the Berka dataset from Kaggle as CSV files covering all eight tables.
2. Restructured card into a JSON file and loan into an Excel workbook, and separated district out from the rest of the CSV export, to create four genuinely distinct sources for this task, while keeping the underlying values unchanged from the original CSV export throughout.
3. Created a dedicated SQL Server database (BerkaSourceDB) and wrote explicit CREATE TABLE scripts for each table using T-SQL data types, based on the data dictionary, rather than relying on an import wizard to infer types.
4. Loaded the five Source 1 CSV tables (account, client, disp, order, trans) into SQL Server using BULK INSERT.
5. Loaded district from Source 4 using a separate BULK INSERT, keeping it distinct from the Source 1 load.
6. Loaded card from card.json using SQL Server's native JSON support (OPENROWSET and OPENJSON), since it is a different extraction method to a plain CSV load.
7. Loaded loan from loan.xlsx using Python and pandas, since SQL Server has no native Excel reader.
8. Verified table and row counts across all four restructured sources against the expected Berka counts, confirming the values are unchanged regardless of file format or source grouping.
9. Set the district name and region columns to NVARCHAR rather than VARCHAR, and confirmed on import that the Czech diacritics in district.csv were preserved correctly.
10. Checked the minimum and maximum dates in trans, order and loan in SQL Server to determine the exact year range the holiday data needs to cover.
11. Retained all data in its original state otherwise. Cleansing tasks such as missing value handling, deduplication and standardisation are performed downstream in the ETL stage (Task 5) and documented there rather than here.

### Tables

#### account

|    | account_id | district_id | frequency        | date       |
|----|------------|-------------|------------------|------------|
| 1  | 1          | 18          | POPLATEK MESICNE | 1995-03-24 |
| 2  | 2          | 1           | POPLATEK MESICNE | 1993-02-26 |
| 3  | 3          | 5           | POPLATEK MESICNE | 1997-07-07 |
| 4  | 4          | 12          | POPLATEK MESICNE | 1996-02-21 |
| 5  | 5          | 15          | POPLATEK MESICNE | 1997-05-30 |
| 6  | 6          | 51          | POPLATEK MESICNE | 1994-09-27 |
| 7  | 7          | 60          | POPLATEK MESICNE | 1996-11-24 |
| 8  | 8          | 57          | POPLATEK MESICNE | 1995-09-21 |
| 9  | 9          | 70          | POPLATEK MESICNE | 1993-01-27 |
| 10 | 10         | 54          | POPLATEK MESICNE | 1996-08-28 |
| 11 | 11         | 76          | POPLATEK MESICNE | 1995-10-10 |
| 12 | 12         | 21          | POPLATEK MESICNE | 1997-04-15 |
| 13 | 13         | 76          | POPLATEK MESICNE | 1997-08-17 |
| 14 | 14         | 47          | POPLATEK MESICNE | 1996-11-27 |
| 15 | 15         | 70          | POPLATEK MESICNE | 1993-10-02 |
| 16 | 16         | 12          | POPLATEK MESICNE | 1997-09-23 |
| 17 | 17         | 1           | POPLATEK MESICNE | 1997-01-08 |

Query executed successfully.

## card

|    | card_id | disp_id | card_type | issued              |
|----|---------|---------|-----------|---------------------|
| 1  | 1       | 9       | gold      | 1998-10-16 00:00:00 |
| 2  | 2       | 19      | classic   | 1998-03-13 00:00:00 |
| 3  | 3       | 41      | gold      | 1995-09-03 00:00:00 |
| 4  | 4       | 42      | classic   | 1998-11-26 00:00:00 |
| 5  | 5       | 51      | junior    | 1995-04-24 00:00:00 |
| 6  | 7       | 56      | classic   | 1998-06-11 00:00:00 |
| 7  | 8       | 60      | junior    | 1998-05-20 00:00:00 |
| 8  | 9       | 76      | classic   | 1997-10-25 00:00:00 |
| 9  | 10      | 77      | classic   | 1996-12-07 00:00:00 |
| 10 | 11      | 79      | gold      | 1997-10-25 00:00:00 |
| 11 | 12      | 83      | junior    | 1996-09-11 00:00:00 |
| 12 | 13      | 87      | classic   | 1994-06-29 00:00:00 |
| 13 | 14      | 112     | classic   | 1996-02-17 00:00:00 |
| 14 | 15      | 114     | classic   | 1995-03-05 00:00:00 |
| 15 | 16      | 116     | classic   | 1998-06-23 00:00:00 |
| 16 | 17      | 127     | classic   | 1998-06-07 00:00:00 |
| 17 | 18      | 128     | classic   | 1995-08-25 00:00:00 |

Query executed successfully.

## client

|    | client_id | birth_number | district_id |
|----|-----------|--------------|-------------|
| 1  | 1         | 1970-12-13   | 18          |
| 2  | 2         | 1945-02-04   | 1           |
| 3  | 3         | 1940-10-09   | 1           |
| 4  | 4         | 1956-12-01   | 5           |
| 5  | 5         | 1960-07-03   | 5           |
| 6  | 6         | 1919-09-22   | 12          |
| 7  | 7         | 1929-01-25   | 15          |
| 8  | 8         | 1938-02-21   | 51          |
| 9  | 9         | 1935-10-16   | 60          |
| 10 | 10        | 1943-05-01   | 57          |
| 11 | 11        | 1950-08-22   | 57          |
| 12 | 12        | 1981-02-20   | 40          |
| 13 | 13        | 1974-05-29   | 54          |
| 14 | 14        | 1942-06-22   | 76          |
| 15 | 15        | 1918-08-28   | 21          |
| 16 | 16        | 1919-02-25   | 21          |
| 17 | 17        | 1934-10-13   | 76          |

Query executed successfully.

## disp

|    | disp_id | client_id | account_id | type      |
|----|---------|-----------|------------|-----------|
| 1  | 1       | 1         | 1          | OWNER     |
| 2  | 2       | 2         | 2          | OWNER     |
| 3  | 3       | 3         | 2          | DISPONENT |
| 4  | 4       | 4         | 3          | OWNER     |
| 5  | 5       | 5         | 3          | DISPONENT |
| 6  | 6       | 6         | 4          | OWNER     |
| 7  | 7       | 7         | 5          | OWNER     |
| 8  | 8       | 8         | 6          | OWNER     |
| 9  | 9       | 9         | 7          | OWNER     |
| 10 | 10      | 10        | 8          | OWNER     |
| 11 | 11      | 11        | 8          | DISPONENT |
| 12 | 12      | 12        | 9          | OWNER     |
| 13 | 13      | 13        | 10         | OWNER     |
| 14 | 14      | 14        | 11         | OWNER     |
| 15 | 15      | 15        | 12         | OWNER     |
| 16 | 16      | 16        | 12         | DISPONENT |
| 17 | 17      | 17        | 13         | OWNER     |

Query executed successfully.

## District

| district_id | district_name | region_name      | population | municipalities_under_500 | municipalities_500_to_1999 | municipalities_2000_to_9999 | municipalities_10000_or_more | number_of_cities | urban_population_percentage |
|-------------|---------------|------------------|------------|--------------------------|----------------------------|-----------------------------|------------------------------|------------------|-----------------------------|
| 1           | 1             | Hl.m. Praha      | 1204953    | 0                        | 0                          | 0                           | 1                            | 1                | 100                         |
| 2           | 2             | Benesov          | 88884      | 80                       | 26                         | 6                           | 2                            | 5                | 46.7000007629395            |
| 3           | 3             | Beroun           | 75232      | 55                       | 26                         | 4                           | 1                            | 5                | 41.7000007629395            |
| 4           | 4             | Kladno           | 149893     | 63                       | 29                         | 6                           | 2                            | 6                | 67.4000015258789            |
| 5           | 5             | Kolin            | 95616      | 65                       | 30                         | 4                           | 1                            | 6                | 51.4000015258789            |
| 6           | 6             | Kutna Hora       | 77963      | 60                       | 23                         | 4                           | 2                            | 4                | 51.5                        |
| 7           | 7             | Melnik           | 94725      | 38                       | 28                         | 1                           | 3                            | 6                | 63.4000015258789            |
| 8           | 8             | Mlada Boleslav   | 112065     | 95                       | 19                         | 7                           | 1                            | 8                | 69.4000015258789            |
| 9           | 9             | Nymburk          | 81344      | 61                       | 23                         | 4                           | 2                            | 6                | 55.2999992370605            |
| 10          | 10            | Praha - vychod   | 92084      | 55                       | 29                         | 4                           | 3                            | 5                | 46.7000007629395            |
| 11          | 11            | Praha - zapad    | 75637      | 35                       | 36                         | 9                           | 0                            | 7                | 36.5                        |
| 12          | 12            | Pribram          | 107870     | 84                       | 29                         | 6                           | 1                            | 6                | 58                          |
| 13          | 13            | Rakovnik         | 53921      | 61                       | 22                         | 1                           | 1                            | 2                | 41.2999992370605            |
| 14          | 14            | Ceske Budejovice | 177686     | 69                       | 27                         | 10                          | 1                            | 9                | 74.8000030517578            |
| 15          | 15            | Cesky Krumlov    | 58796      | 22                       | 16                         | 7                           | 1                            | 5                | 51.9000015258789            |
| 16          | 16            | Stredocesky kraj | 62021      | 74                       | 21                         | 10                          | 1                            | 6                | 56.0000015258789            |

| alities_10000_or_more | number_of_cities | urban_population_percentage | average_salary | unemployment_rate_1995 | unemployment_rate_1996 | entrepreneurs_per_1000_residents | crimes_committed_1995 | crimes_committed_1996 |
|-----------------------|------------------|-----------------------------|----------------|------------------------|------------------------|----------------------------------|-----------------------|-----------------------|
| 1                     | 1                | 100                         | 12541          | 0.28999999165535       | 0.430000007152557      | 167                              | 85677                 | 99107                 |
| 2                     | 5                | 46.7000007629395            | 8507           | 1.66999995708466       | 1.85000002384186       | 132                              | 2159                  | 2674                  |
| 3                     | 5                | 41.7000007629395            | 8980           | 1.95000004768372       | 2.21000003814697       | 111                              | 2824                  | 2813                  |
| 4                     | 6                | 67.4000015258789            | 9753           | 4.6399998664856        | 5.05000019073486       | 109                              | 5244                  | 5892                  |
| 5                     | 6                | 51.4000015258789            | 9307           | 3.84999990463257       | 4.42999982833862       | 118                              | 2616                  | 3040                  |
| 6                     | 4                | 51.5                        | 8546           | 2.95000004768372       | 4.01999998092651       | 126                              | 2640                  | 3120                  |
| 7                     | 6                | 63.4000015258789            | 9920           | 2.2599999046326        | 2.86999988555908       | 130                              | 4289                  | 4846                  |
| 8                     | 8                | 69.4000015258789            | 11277          | 1.25                   | 1.44000005722046       | 127                              | 5179                  | 4987                  |
| 9                     | 6                | 55.2999992370605            | 8899           | 3.39000010490417       | 3.97000002861023       | 149                              | 2987                  | 2487                  |
| 10                    | 5                | 46.7000007629395            | 10124          | 0.560000002384186      | 0.540000021457672      | 141                              | 3810                  | 4316                  |
| 11                    | 7                | 36.5                        | 9622           | 0.449999988079071      | 0.589999973773956      | 154                              | 3475                  | 3529                  |
| 12                    | 6                | 58                          | 8754           | 3.82999992370605       | 4.30999994277954       | 137                              | 3804                  | 3868                  |
| 13                    | 2                | 41.2999992370605            | 8598           | 2.76999998092651       | 3.25999999046326       | 123                              | 1597                  | 1875                  |
| 14                    | 9                | 74.8000030517578            | 10045          | 1.41999995708466       | 1.71000003814697       | 135                              | 8604                  | 6295                  |

Query executed successfully. DESKTOP-FSAFHJV (17.0 RTM) DESKTOP-FSAFHJV\... CzechBank\_DW 00:00:00 Row: 1, Col: 1 77 rows

## Loan

Results

Messages

|    | loan_id | account_id | date       | amount | duration | payments | status |
|----|---------|------------|------------|--------|----------|----------|--------|
| 1  | 4959    | 2          | 1994-01-05 | 80952  | 24       | 3373     | A      |
| 2  | 4961    | 19         | 1996-04-29 | 30276  | 12       | 2523     | B      |
| 3  | 4962    | 25         | 1997-12-08 | 30276  | 12       | 2523     | A      |
| 4  | 4967    | 37         | 1998-10-14 | 318480 | 60       | 5308     | D      |
| 5  | 4968    | 38         | 1998-04-19 | 110736 | 48       | 2307     | C      |
| 6  | 4973    | 67         | 1996-05-02 | 165960 | 24       | 6915     | A      |
| 7  | 4986    | 97         | 1997-08-10 | 102876 | 12       | 8573     | A      |
| 8  | 4988    | 103        | 1997-12-06 | 265320 | 36       | 7370     | D      |
| 9  | 4989    | 105        | 1998-12-05 | 352704 | 48       | 7348     | C      |
| 10 | 4990    | 110        | 1997-09-08 | 162576 | 36       | 4516     | C      |
| 11 | 4996    | 132        | 1996-11-06 | 88440  | 12       | 7370     | A      |
| 12 | 5002    | 173        | 1994-05-31 | 104808 | 12       | 8734     | A      |
| 13 | 5005    | 176        | 1997-04-10 | 27456  | 12       | 2288     | A      |
| 14 | 5015    | 226        | 1998-07-12 | 109344 | 12       | 9112     | C      |
| 15 | 5027    | 276        | 1998-12-02 | 160920 | 36       | 4470     | C      |
| 16 | 5032    | 290        | 1994-11-03 | 123696 | 48       | 2577     | A      |

## Order

|    | order_id | account_id | bank_to | account_to | amount           | k_symbol |
|----|----------|------------|---------|------------|------------------|----------|
| 1  | 29401    | 1          | YZ      | 87144583   | 2452             | SIPO     |
| 2  | 29402    | 2          | ST      | 89597016   | 3372.69995117188 | UVER     |
| 3  | 29403    | 2          | QR      | 13943797   | 7266             | SIPO     |
| 4  | 29404    | 3          | WX      | 83084338   | 1135             | SIPO     |
| 5  | 29405    | 3          | CD      | 24485939   | 327              |          |
| 6  | 29406    | 3          | AB      | 59972357   | 3539             | POJISTNE |
| 7  | 29407    | 4          | UV      | 26693541   | 2078             | SIPO     |
| 8  | 29408    | 4          | UV      | 5848086    | 1285             | SIPO     |
| 9  | 29409    | 5          | GH      | 37390208   | 2668             | SIPO     |
| 10 | 29410    | 6          | AB      | 44486999   | 3954             | SIPO     |
| 11 | 29411    | 7          | IJ      | 34452903   | 4880             | SIPO     |
| 12 | 29412    | 8          | UV      | 12510681   | 2612             | SIPO     |
| 13 | 29413    | 8          | IJ      | 93210345   | 6712             |          |
| 14 | 29414    | 10         | UV      | 18686104   | 7033             | SIPO     |
| 15 | 29415    | 10         | QR      | 93182509   | 1344             | LEASING  |
| 16 | 29416    | 11         | ST      | 38470870   | 2132             | SIPO     |

## Transaction

|    | trans_id | account_id | date       | type   | operation     | amount | balance         | k_symbol | bank | account  |
|----|----------|------------|------------|--------|---------------|--------|-----------------|----------|------|----------|
| 1  | 1        | 1          | 1995-03-24 | PRIJEM | VKLAD         | 1000   | 1000            | NULL     | NULL | NULL     |
| 2  | 5        | 1          | 1995-04-13 | PRIJEM | PREVOD Z UCTU | 3679   | 4679            | NULL     | AB   | 41403269 |
| 3  | 6        | 1          | 1995-05-13 | PRIJEM | PREVOD Z UCTU | 3679   | 20977.19921875  | NULL     | AB   | 41403269 |
| 4  | 7        | 1          | 1995-06-13 | PRIJEM | PREVOD Z UCTU | 3679   | 26835.19921875  | NULL     | AB   | 41403269 |
| 5  | 8        | 1          | 1995-07-13 | PRIJEM | PREVOD Z UCTU | 3679   | 30414.80078125  | NULL     | AB   | 41403269 |
| 6  | 9        | 1          | 1995-08-13 | PRIJEM | PREVOD Z UCTU | 3679   | 28902.69921875  | NULL     | AB   | 41403269 |
| 7  | 10       | 1          | 1995-09-13 | PRIJEM | PREVOD Z UCTU | 3679   | 22714.30078125  | NULL     | AB   | 41403269 |
| 8  | 11       | 1          | 1995-10-13 | PRIJEM | PREVOD Z UCTU | 3679   | 23318.19921875  | NULL     | AB   | 41403269 |
| 9  | 12       | 1          | 1995-11-13 | PRIJEM | PREVOD Z UCTU | 3679   | 21721.099609375 | NULL     | AB   | 41403269 |
| 10 | 13       | 1          | 1995-12-13 | PRIJEM | PREVOD Z UCTU | 3679   | 20248.599609375 | NULL     | AB   | 41403269 |
| 11 | 14       | 1          | 1996-01-13 | PRIJEM | PREVOD Z UCTU | 3679   | 17870           | NULL     | AB   | 41403269 |
| 12 | 15       | 1          | 1996-02-13 | PRIJEM | PREVOD Z UCTU | 3679   | 14806.5         | NULL     | AB   | 41403269 |
| 13 | 16       | 1          | 1996-03-13 | PRIJEM | PREVOD Z UCTU | 3679   | 16178.599609375 | NULL     | AB   | 41403269 |
| 14 | 17       | 1          | 1996-04-13 | PRIJEM | PREVOD Z UCTU | 3679   | 17656.80078125  | NULL     | AB   | 41403269 |
| 15 | 18       | 1          | 1996-05-13 | PRIJEM | PREVOD Z UCTU | 3679   | 16936.69921875  | NULL     | AB   | 41403269 |

Query executed successfully. DESKTOP-FSAFHJV (17.0 RTM) DESKTOP-FSAFHJV\nuwin ... Cz

## Integration into a Unified Analytical Environment

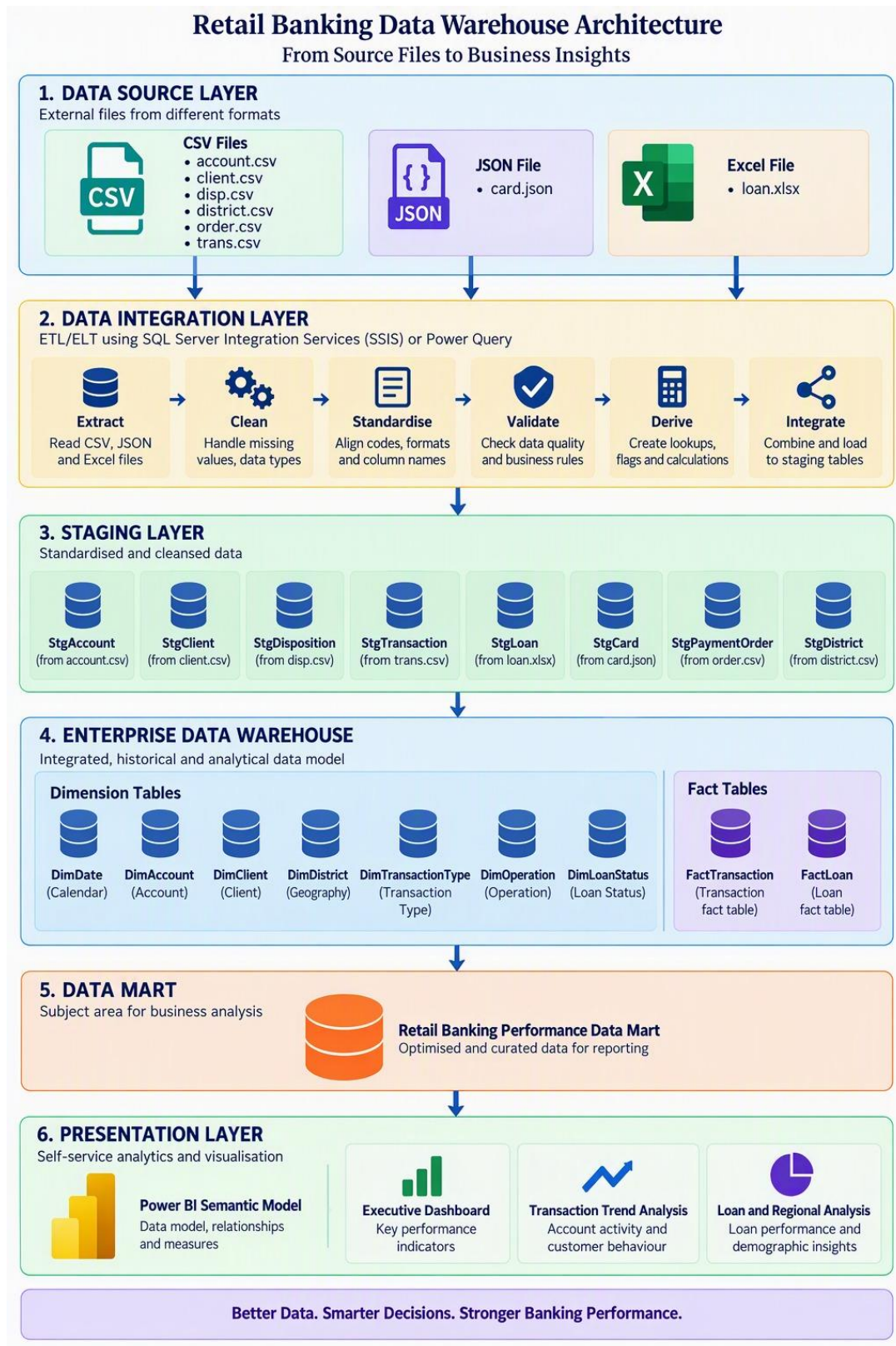
Although the five sources differ in format or grouping, four of them share not only a business domain but the exact same underlying values, since they all originate from the same Kaggle download restructured or regrouped by the team. Integration is therefore straightforward for Sources 1 to 4: each is extracted in its own way, a CSV read for Sources 1 and 4, a JSON parse for Source 3, and an Excel read for Source 2, then loaded into the same SQL Server staging database, where the original account\_id, disp\_id and district\_id keys reconnect them exactly as they would if all four were still one CSV export.

## Member 2 Deliverable: Tasks 3 and 4

Prepared by: Welgama W. N. Y (Member 2)

### Task 3: Data Warehouse Architecture

#### 3.1 Proposed architecture



## 3.2 Component explanation

### Data source layer

We use [1999 Czech Financial Dataset in Kaggle](#) and all files are csv but we convert it to different formats. Such as .xlsx, .json files.

### Data integration layer

Extracts data from each source files and applies cleansing, standardization, validation and business transformations.

### Staging layer

Temporarily stores source data without changing the original operational system. It also supports validation and troubleshooting.

### Data Warehouse layer

Stores integrated historical information in dimensional form. Surrogate keys are used to improve dimension management and isolate the warehouse from operational keys.

### Data mart layer

Provides subject-specific transaction and loan information for managers and financial analysts.

### Presentation layer

Power BI connects to the dimensional model and provides KPIs, visualizations, slicers, drill-down and roll-up.

## Task 4: Dimensional Data Warehouse Design

### 4.1 Recommended approach

Use a fact constellation, also called a galaxy schema, with two fact tables:

1. FactLoan

This is better than forcing transactions and loans into one fact table because they have different grains and measures.

2. FactTransaction

## 4.2 FactTransaction

### Business process

Recording money moving into or out of a bank account.

### Grain

One row represents one individual transaction recorded against one bank account.

### Foreign keys

- *DateKey*
- *AccountKey*
- *DistrictKey*
- *TransactionTypeKey*
- *OperationKey*

### Degenerate dimension

- *TransactionID*

### Measures

- *TransactionCount*
- *TransactionAmount*
- *InflowAmount*
- *OutflowAmount*
- *SignedAmount*
- *BalanceAfterTransaction*

|    | TransactionKey | TransactionID | DateKey  | AccountKey | DistrictKey | TransactionTypeKey | OperationKey | TransactionCount | TransactionAmount | InflowAmount | OutflowAmount | SignedAmount | BalanceAfterTransaction |
|----|----------------|---------------|----------|------------|-------------|--------------------|--------------|------------------|-------------------|--------------|---------------|--------------|-------------------------|
| 1  | 2112641        | 200263        | 19980806 | 4104       | 17          | 1                  | 3            | 1                | 14333.00          | 14333.00     | 0.00          | 14333.00     | 47177.00                |
| 2  | 2112642        | 200264        | 19980906 | 4104       | 17          | 1                  | 3            | 1                | 14333.00          | 14333.00     | 0.00          | 14333.00     | 45376.50                |
| 3  | 2112643        | 200302        | 19980105 | 4104       | 17          | 3                  | 4            | 1                | 2280.00           | 0.00         | 2280.00       | -2280.00     | 40103.20                |
| 4  | 2112644        | 200303        | 19980113 | 4104       | 17          | 3                  | 4            | 1                | 2280.00           | 0.00         | 2280.00       | -2280.00     | 44423.20                |
| 5  | 2112645        | 200304        | 19980102 | 4104       | 17          | 3                  | 4            | 1                | 1800.00           | 0.00         | 1800.00       | -1800.00     | 44663.20                |
| 6  | 2112646        | 200307        | 19980118 | 4104       | 17          | 3                  | 4            | 1                | 600.00            | 0.00         | 600.00        | -600.00      | 41223.20                |
| 7  | 2112647        | 200311        | 19980404 | 4104       | 17          | 3                  | 4            | 1                | 2400.00           | 0.00         | 2400.00       | -2400.00     | 45418.80                |
| 8  | 2112648        | 200327        | 19980228 | 4104       | 17          | 3                  | 4            | 1                | 14.60             | 0.00         | 14.60         | -14.60       | 47049.70                |
| 9  | 2112649        | 200333        | 19980831 | 4104       | 17          | 3                  | 4            | 1                | 14.60             | 0.00         | 14.60         | -14.60       | 31043.50                |
| 10 | 2112650        | 200343        | 19980213 | 4104       | 17          | 3                  | 4            | 1                | 4000.00           | 0.00         | 4000.00       | -4000.00     | 46874.50                |
| 11 | 2112651        | 200347        | 19980613 | 4104       | 17          | 1                  | 3            | 1                | 4500.00           | 4500.00      | 0.00          | 4500.00      | 27937.50                |
| 12 | 2112652        | 200348        | 19980713 | 4104       | 17          | 3                  | 4            | 1                | 4800.00           | 0.00         | 4800.00       | -4800.00     | 32726.10                |
| 13 | 2112653        | 286           | 19931012 | 182        | 1           | 1                  | 2            | 1                | 20236.00          | 20236.00     | 0.00          | 20236.00     | 30744.00                |
| 14 | 2112654        | 288           | 19931212 | 182        | 1           | 1                  | 2            | 1                | 30354.00          | 30354.00     | 0.00          | 30354.00     | 47212.90                |
| 15 | 2112655        | 289           | 19940112 | 182        | 1           | 1                  | 2            | 1                | 20236.00          | 20236.00     | 0.00          | 20236.00     | 40810.60                |

## 4.3 FactLoan

### Business process

Monitoring loans issued to bank accounts.

### Grain

One row represents one loan issued to one bank account.

## Foreign keys

- *LoanDateKey*
- *AccountKey*
- *DistrictKey*
- *LoanStatusKey*

## Measures

- *LoanCount*
- *LoanAmount*
- *LoanDurationMonths*
- *MonthlyPayment*
- *ExpectedPaymentAmount*
- *ProblemLoanCount*

|    | LoanKey | LoanID | LoanDateKey | AccountKey | DistrictKey | LoanStatusKey | LoanCount | LoanAmount | LoanDurationMonths | MonthlyPayment | ExpectedPaymentAmount | ProblemLoanCount |
|----|---------|--------|-------------|------------|-------------|---------------|-----------|------------|--------------------|----------------|-----------------------|------------------|
| 1  | 1       | 4959   | 19940105    | 182        | 1           | 1             | 1         | 80952.00   | 24                 | 3373.00        | 80952.00              | 0                |
| 2  | 2       | 4961   | 19960429    | 1751       | 21          | 2             | 1         | 30276.00   | 12                 | 2523.00        | 30276.00              | 1                |
| 3  | 3       | 4962   | 19971208    | 3003       | 68          | 1             | 1         | 30276.00   | 12                 | 2523.00        | 30276.00              | 0                |
| 4  | 4       | 4967   | 19981014    | 4187       | 20          | 4             | 1         | 318480.00  | 60                 | 5308.00        | 318480.00             | 1                |
| 5  | 5       | 4968   | 19980419    | 4159       | 19          | 3             | 1         | 110736.00  | 48                 | 2307.00        | 110736.00             | 0                |
| 6  | 6       | 4973   | 19960502    | 1502       | 16          | 1             | 1         | 165960.00  | 24                 | 6915.00        | 165960.00             | 0                |
| 7  | 7       | 4986   | 19970810    | 2687       | 74          | 1             | 1         | 102876.00  | 12                 | 8573.00        | 102876.00             | 0                |
| 8  | 8       | 4988   | 19971206    | 2486       | 44          | 4             | 1         | 265320.00  | 36                 | 7370.00        | 265320.00             | 1                |
| 9  | 9       | 4989   | 19981205    | 4082       | 21          | 3             | 1         | 352704.00  | 48                 | 7348.00        | 352704.00             | 0                |
| 10 | 10      | 4990   | 19970908    | 2958       | 36          | 3             | 1         | 162576.00  | 36                 | 4516.00        | 162576.00             | 0                |
| 11 | 11      | 4996   | 19961106    | 2713       | 40          | 1             | 1         | 88440.00   | 12                 | 7370.00        | 88440.00              | 0                |
| 12 | 12      | 5002   | 19940531    | 1030       | 66          | 1             | 1         | 104808.00  | 12                 | 8734.00        | 104808.00             | 0                |
| 13 | 13      | 5005   | 19970410    | 3061       | 71          | 1             | 1         | 27456.00   | 12                 | 2288.00        | 27456.00              | 0                |
| 14 | 14      | 5015   | 19980712    | 3730       | 70          | 3             | 1         | 109344.00  | 12                 | 9112.00        | 109344.00             | 0                |
| 15 | 15      | 5027   | 19981202    | 4454       | 38          | 3             | 1         | 160920.00  | 36                 | 4470.00        | 160920.00             | 0                |
| 16 | 16      | 5032   | 19941103    | 1052       | 56          | 1             | 1         | 123696.00  | 48                 | 2577.00        | 123696.00             | 0                |

## 4.4 Dimensions

- **DimDate:** (*DateKey*, FullDate, DayNumber, DayName, WeekNumber, MonthNumber, MonthName, Quarter, Year, YearMonth)

|    | DateKey  | FullDate   | DayNumber | DayName   | WeekNumber | MonthNumber | MonthName | Quarter | Year | YearMonth |
|----|----------|------------|-----------|-----------|------------|-------------|-----------|---------|------|-----------|
| 1  | 19000101 | 1900-01-01 | 1         | Monday    | 1          | 1           | January   | 1       | 1900 | 01        |
| 2  | 19000102 | 1900-01-02 | 2         | Tuesday   | 1          | 1           | January   | 1       | 1900 | 01        |
| 3  | 19000103 | 1900-01-03 | 3         | Wednesday | 1          | 1           | January   | 1       | 1900 | 01        |
| 4  | 19000104 | 1900-01-04 | 4         | Thursday  | 1          | 1           | January   | 1       | 1900 | 01        |
| 5  | 19000105 | 1900-01-05 | 5         | Friday    | 1          | 1           | January   | 1       | 1900 | 01        |
| 6  | 19000106 | 1900-01-06 | 6         | Saturday  | 1          | 1           | January   | 1       | 1900 | 01        |
| 7  | 19000107 | 1900-01-07 | 7         | Sunday    | 1          | 1           | January   | 1       | 1900 | 01        |
| 8  | 19000108 | 1900-01-08 | 8         | Monday    | 2          | 1           | January   | 1       | 1900 | 01        |
| 9  | 19000109 | 1900-01-09 | 9         | Tuesday   | 2          | 1           | January   | 1       | 1900 | 01        |
| 10 | 19000110 | 1900-01-10 | 10        | Wednesday | 2          | 1           | January   | 1       | 1900 | 01        |
| 11 | 19000111 | 1900-01-11 | 11        | Thursday  | 2          | 1           | January   | 1       | 1900 | 01        |
| 12 | 19000112 | 1900-01-12 | 12        | Friday    | 2          | 1           | January   | 1       | 1900 | 01        |
| 13 | 19000113 | 1900-01-13 | 13        | Saturday  | 2          | 1           | January   | 1       | 1900 | 01        |
| 14 | 19000114 | 1900-01-14 | 14        | Sunday    | 2          | 1           | January   | 1       | 1900 | 01        |
| 15 | 19000115 | 1900-01-15 | 15        | Monday    | 3          | 1           | January   | 1       | 1900 | 01        |
| 16 | 19000116 | 1900-01-16 | 16        | Tuesday   | 3          | 1           | January   | 1       | 1900 | 01        |
| 17 | 19000117 | 1900-01-17 | 17        | Wednesday | 3          | 1           | January   | 1       | 1900 | 01        |

- **DimAccount:** (AccountKey, AccountID, AccountOpenDate, StatementFrequency, AccountAgeGroup, SourceSystem, EffectiveFrom, EffectiveTo, IsCurrent)

| Results |            | Messages  |                 |                    |                 |              |               |             |           |
|---------|------------|-----------|-----------------|--------------------|-----------------|--------------|---------------|-------------|-----------|
|         | AccountKey | AccountID | AccountOpenDate | StatementFrequency | AccountAgeGroup | SourceSystem | EffectiveFrom | EffectiveTo | IsCurrent |
| 1       | 0          | -1        | 19000101        | Unknown            | Unknown         | Unknown      | 1900-01-01    | 9999-12-31  | 1         |
| 2       | 1          | 576       | 19930101        | Monthly            | 5-9 Years       | CzechBank_DW | 1993-01-01    | 9999-12-31  | 1         |
| 3       | 2          | 704       | 19930101        | Monthly            | 5-9 Years       | CzechBank_DW | 1993-01-01    | 9999-12-31  | 1         |
| 4       | 3          | 2378      | 19930101        | Monthly            | 5-9 Years       | CzechBank_DW | 1993-01-01    | 9999-12-31  | 1         |
| 5       | 4          | 3818      | 19930101        | Monthly            | 5-9 Years       | CzechBank_DW | 1993-01-01    | 9999-12-31  | 1         |
| 6       | 5          | 2632      | 19930102        | Monthly            | 5-9 Years       | CzechBank_DW | 1993-01-02    | 9999-12-31  | 1         |
| 7       | 6          | 1972      | 19930102        | Monthly            | 5-9 Years       | CzechBank_DW | 1993-01-02    | 9999-12-31  | 1         |
| 8       | 7          | 1539      | 19930103        | After Transaction  | 5-9 Years       | CzechBank_DW | 1993-01-03    | 9999-12-31  | 1         |
| 9       | 8          | 1695      | 19930103        | Monthly            | 5-9 Years       | CzechBank_DW | 1993-01-03    | 9999-12-31  | 1         |
| 10      | 9          | 1726      | 19930103        | Monthly            | 5-9 Years       | CzechBank_DW | 1993-01-03    | 9999-12-31  | 1         |
| 11      | 10         | 793       | 19930103        | Monthly            | 5-9 Years       | CzechBank_DW | 1993-01-03    | 9999-12-31  | 1         |
| 12      | 11         | 2484      | 19930103        | Monthly            | 5-9 Years       | CzechBank_DW | 1993-01-03    | 9999-12-31  | 1         |
| 13      | 12         | 2881      | 19930104        | Monthly            | 5-9 Years       | CzechBank_DW | 1993-01-04    | 9999-12-31  | 1         |
| 14      | 13         | 485       | 19930104        | After Transaction  | 5-9 Years       | CzechBank_DW | 1993-01-04    | 9999-12-31  | 1         |
| 15      | 14         | 2357      | 19930104        | Monthly            | 5-9 Years       | CzechBank_DW | 1993-01-04    | 9999-12-31  | 1         |
| 16      | 15         | 2177      | 19930104        | Monthly            | 5-9 Years       | CzechBank_DW | 1993-01-04    | 9999-12-31  | 1         |
| 17      | 16         | 652       | 19930105        | Monthly            | 5-9 Years       | CzechBank_DW | 1993-01-05    | 9999-12-31  | 1         |

- **DimClient:** (ClientKey, ClientID, DistrictKey, ClientRole, EffectiveFrom, EffectiveTo, IsCurrent)

Results

Messages

|    | ClientKey | ClientID | DistrictKey | ClientRole      | EffectiveFrom | EffectiveTo | IsCurrent |
|----|-----------|----------|-------------|-----------------|---------------|-------------|-----------|
| 1  | 1         | 1        | 18          | Owner           | 1995-03-24    | 9999-12-31  | 1         |
| 2  | 2         | 2        | 1           | Owner           | 1993-02-26    | 9999-12-31  | 1         |
| 3  | 3         | 3        | 1           | Authorized User | 1993-02-26    | 9999-12-31  | 1         |
| 4  | 4         | 4        | 5           | Owner           | 1997-07-07    | 9999-12-31  | 1         |
| 5  | 5         | 5        | 5           | Authorized User | 1997-07-07    | 9999-12-31  | 1         |
| 6  | 6         | 6        | 12          | Owner           | 1996-02-21    | 9999-12-31  | 1         |
| 7  | 7         | 7        | 15          | Owner           | 1997-05-30    | 9999-12-31  | 1         |
| 8  | 8         | 8        | 51          | Owner           | 1994-09-27    | 9999-12-31  | 1         |
| 9  | 9         | 9        | 60          | Owner           | 1996-11-24    | 9999-12-31  | 1         |
| 10 | 10        | 10       | 57          | Owner           | 1995-09-21    | 9999-12-31  | 1         |
| 11 | 11        | 11       | 57          | Authorized User | 1995-09-21    | 9999-12-31  | 1         |
| 12 | 12        | 12       | 40          | Owner           | 1993-01-27    | 9999-12-31  | 1         |
| 13 | 13        | 13       | 54          | Owner           | 1996-08-28    | 9999-12-31  | 1         |
| 14 | 14        | 14       | 76          | Owner           | 1995-10-10    | 9999-12-31  | 1         |
| 15 | 15        | 15       | 21          | Owner           | 1997-04-15    | 9999-12-31  | 1         |
| 16 | 16        | 16       | 21          | Authorized User | 1997-04-15    | 9999-12-31  | 1         |
| 17 | 17        | 17       | 76          | Owner           | 1997-08-17    | 9999-12-31  | 1         |

- **DimDistrict:** (DistrictKey, DistrictID, DistrictName, RegionName, Population, AverageSalary, UnemploymentMeasure, EntrepreneurMeasure, CrimeMeasure)

|    | DistrictKey | DistrictID | DistrictName      | RegionName      | Population | AverageSalary | UnemploymentMeasure | EntrepreneurMeasure | CrimeMeasure |
|----|-------------|------------|-------------------|-----------------|------------|---------------|---------------------|---------------------|--------------|
| 1  | 0           | -1         | Unknown           | Unknown         | NULL       | NULL          | NULL                | NULL                | NULL         |
| 2  | 1           | 1          | Hl.m. Praha       | Prague          | 1204953    | 12541         | 0.43                | 167                 | 99107        |
| 3  | 2           | 2          | Benesov           | central Bohemia | 88884      | 8507          | 1.85                | 132                 | 2674         |
| 4  | 3           | 3          | Beroun            | central Bohemia | 75232      | 8980          | 2.21                | 111                 | 2813         |
| 5  | 4           | 4          | Kladno            | central Bohemia | 149893     | 9753          | 5.05                | 109                 | 5892         |
| 6  | 5           | 5          | Kolin             | central Bohemia | 95616      | 9307          | 4.43                | 118                 | 3040         |
| 7  | 6           | 6          | Kutna Hora        | central Bohemia | 77963      | 8546          | 4.02                | 126                 | 3120         |
| 8  | 7           | 7          | Melnik            | central Bohemia | 94725      | 9920          | 2.87                | 130                 | 4846         |
| 9  | 8           | 8          | Mlada Boleslav    | central Bohemia | 112065     | 11277         | 1.44                | 127                 | 4987         |
| 10 | 9           | 9          | Nymburk           | central Bohemia | 81344      | 8899          | 3.97                | 149                 | 2487         |
| 11 | 10          | 10         | Praha - vychod    | central Bohemia | 92084      | 10124         | 0.54                | 141                 | 4316         |
| 12 | 11          | 11         | Praha - zapad     | central Bohemia | 75637      | 9622          | 0.59                | 154                 | 3529         |
| 13 | 12          | 12         | Pribram           | central Bohemia | 107870     | 8754          | 4.31                | 137                 | 3868         |
| 14 | 13          | 13         | Rakovnik          | central Bohemia | 53921      | 8598          | 3.26                | 123                 | 1875         |
| 15 | 14          | 14         | Ceske Budejovice  | south Bohemia   | 177686     | 10045         | 1.71                | 135                 | 6295         |
| 16 | 15          | 15         | Cesky Krumlov     | south Bohemia   | 58796      | 9045          | 3.60                | 124                 | 1879         |
| 17 | 16          | 16         | Jindrichuv Hradec | south Bohemia   | 92031      | 8477          | 1.54                | 107                 | 1013         |

Query executed successfully

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- **DimTransactionType:** (TransactionTypeKey, TransactionTypeCode, TransactionTypeName, CashFlowDirection)

|   | TransactionTypeKey | TransactionTypeCode | TransactionTypeName      | CashFlowDirection |
|---|--------------------|---------------------|--------------------------|-------------------|
| 1 | 0                  | UNKNOWN             | Unknown Transaction Type | Unknown           |
| 2 | 1                  | PRIJEM              | Credit                   | Incoming          |
| 3 | 2                  | VYBER               | Withdrawal               | Outgoing          |
| 4 | 3                  | VYDAJ               | Debit                    | Outgoing          |

- **DimOperation:** (OperationKey, OperationCode, OperationDescription, OperationGroup)

|   | OperationKey | OperationCode  | OperationDescription   | OperationGroup   |
|---|--------------|----------------|------------------------|------------------|
| 1 | 0            | UNKNOWN        | Unknown Operation      | Unknown          |
| 2 | 1            | PREVOD NA UCET | Outgoing Bank Transfer | Bank Transfer    |
| 3 | 2            | PREVOD Z UCTU  | Incoming Bank Transfer | Bank Transfer    |
| 4 | 3            | VKLAD          | Cash Deposit           | Cash Transaction |
| 5 | 4            | VYBER          | Cash Withdrawal        | Cash Transaction |
| 6 | 5            | VYBER KARTOU   | Credit Card Withdrawal | Card Transaction |

- **DimLoanStatus:** (LoanStatusKey, LoanStatusCode, LoanStatusDescription, LoanStatusGroup, ProblemFlag)

|   | LoanStatusKey | LoanStatusCode | LoanStatusDescription                     | LoanStatusGroup | ProblemFlag |
|---|---------------|----------------|-------------------------------------------|-----------------|-------------|
| 1 | 0             | UNKNOWN        | Unknown Loan Status                       | Unknown         | 0           |
| 2 | 1             | A              | Completed Without Repayment Problems      | Completed       | 0           |
| 3 | 2             | B              | Completed With Repayment Problems         | Completed       | 1           |
| 4 | 3             | C              | Active Without Current Repayment Problems | Active          | 0           |
| 5 | 4             | D              | Active With Current Debt                  | Active          | 1           |

## 4.5 Design assumptions

- Surrogate integer keys will be used in dimensions.
- Operational IDs will remain as business keys.
- Unknown dimension members will use key 0.
- Financial values will use a fixed decimal data type.
- Transaction and loan facts will remain separate.
- Date will be a role-playing dimension for account, transaction, loan and card dates.
- Client details will not be exposed unnecessarily in reports.
- Loan status analysis will describe portfolio conditions, not recommend decisions about particular individuals.

## Member 3 Deliverable: Tasks 5 and 6

Prepared by: Hannadige A. K. L. F (Member 3)

### Member 3 Deliverable: Task 5 – ETL Process Development

#### 5.1 ETL Pipeline Overview

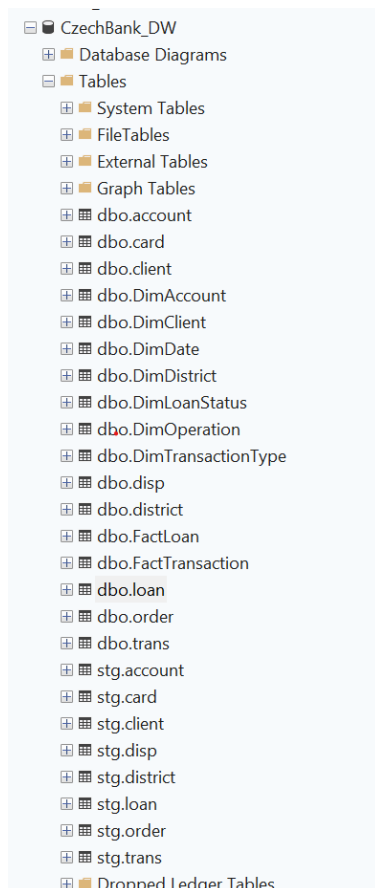
Member 3 implemented the complete ETL flow that moves the prepared banking data from heterogeneous source files into the dimensional warehouse designed by Member 2. The implemented sequence was: Source Files → Raw Staging → Data Profiling → Clean Staging → Dimension Loading → Fact Loading → Data Mart → Final Validation. This separation keeps the original source values auditable while allowing cleansing and business transformations to be applied in a controlled layer.

**Source Files → Raw Staging → Profiling → Clean Staging → Dimensions → Facts → Data Mart**

#### 5.2 Extraction and Raw Staging

All eight logical datasets were brought into SQL Server staging. The source formats included CSV, JSON and Excel. Each staged row received an ETL\_LoadDateTime value for traceability, and source-to-staging row counts were checked before transformation. The original source files were not modified during this process.

| Source dataset | Format        | Raw staging table | Downstream use                                         |
|----------------|---------------|-------------------|--------------------------------------------------------|
| account        | CSV           | stg.account       | DimAccount and account/district lookups                |
| client         | CSV           | stg.client        | DimClient                                              |
| disp           | CSV           | stg.disp          | Integrated with client to derive account role          |
| district       | CSV           | stg.district      | DimDistrict and regional analysis                      |
| order          | CSV           | stg.[order]       | Retained in staging for future standing-order analysis |
| trans          | CSV           | stg.trans         | FactTransaction                                        |
| card           | JSON          | stg.card          | Retained in staging for future card analysis           |
| loan           | Excel (.xlsx) | stg.loan          | FactLoan                                               |



**Figure 1: Raw staging tables created during the extraction stage.**

### 5.3 Data Profiling and Transformation Logic

Profiling was completed before transformation so that cleansing decisions were based on the actual data rather than assumptions. The checks covered column metadata, null values, duplicate business IDs, category distributions, invalid relationships and source-to-dimension mappings. Five clean staging tables were then created for the subjects consumed by the warehouse.

| Clean staging table   | Main transformations                                                                                                                 | Validated result           |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| stg.account_clean     | Standardised statement frequency; created YYYYMMDD open-date key; derived account age and age group; removed duplicate business IDs. | 4,500 rows                 |
| stg.district_clean    | Trimmed names; converted salary to integer; used 1996 unemployment/crime values with 1995 fallback; preserved real economic values.  | 77 rows                    |
| stg.client_clean      | Joined client with disposition; translated OWNER/DISPONENT to English roles; linked account and district; deduplicated.              | 5,369 rows                 |
| stg.loan_clean        | Mapped loan statuses A–D; converted financial values to DECIMAL(18,2); derived expected payments and ProblemLoanCount.               | 682 rows; 76 problem loans |
| stg.transaction_clean | Standardised type/operation codes; created date key; derived inflow, outflow and signed amount; preserved unknown balances as NULL.  | 1,056,320 rows             |

The screenshot shows a SQL query result window with four tables of profiling data. The first table shows frequency counts for three categories. The second table shows record counts for three types. The third table shows record counts for six operations, with 'NULL' highlighted. The fourth table shows record counts for four statuses, with 'A' highlighted.

|   | frequency          | RecordCount |
|---|--------------------|-------------|
| 1 | POPLATEK MESICNE   | 4167        |
| 2 | POPLATEK TYDNE     | 240         |
| 3 | POPLATEK PO OBRATU | 93          |

|   | type   | RecordCount |
|---|--------|-------------|
| 1 | VYDAJ  | 634571      |
| 2 | PRIJEM | 405083      |
| 3 | VYBER  | 16666       |

|   | operation      | RecordCount |
|---|----------------|-------------|
| 1 | VYBER          | 434918      |
| 2 | PREVOD NA UCET | 208283      |
| 3 | NULL           | 183114      |
| 4 | VKLAD          | 156743      |
| 5 | PREVOD Z UCTU  | 65226       |
| 6 | VYBER KARTOU   | 8036        |

|   | status | RecordCount |
|---|--------|-------------|
| 1 | A      | 203         |
| 2 | B      | 31          |
| 3 | C      | 403         |
| 4 | D      | 45          |

**Figure 2: Source value profiling results used to define transformation rules.**

Key data-quality decisions were deliberately conservative so that valid business events were not removed or changed:

- Problem loans with statuses B and D were retained because they are valid events required for portfolio risk analysis.
- Unknown dimension members were preserved so unmatched categories or locations do not cause fact records to be dropped.
- 2,999 missing transaction balances were preserved as NULL rather than replaced with zero, because zero has a real financial meaning.
- 183,114 missing transaction-operation values were mapped to the Unknown Operation dimension member.
- Financial FLOAT values were converted to DECIMAL(18,2) to avoid approximate floating-point values in reporting.
- Original Czech business codes were retained as business keys while English descriptions were provided through lookup dimensions.

## 5.4 Dimension Loading

Dimensions were loaded before facts so that all required surrogate keys existed before foreign-key resolution. The load logic matched existing records by business key, updated standardised attributes where required, and inserted only records that were missing. SQL Server identity columns generated the warehouse surrogate keys.

| Target             | Business-key match / load behaviour                                                                         | Final count             |
|--------------------|-------------------------------------------------------------------------------------------------------------|-------------------------|
| DimDistrict        | Match on DistrictID; update cleaned attributes; insert missing; preserve Unknown member.                    | 78 total (77 + Unknown) |
| DimAccount         | Match on AccountID/current record; update standardised attributes; insert missing; preserve Unknown member. | 4,501 total             |
| DimClient          | Match on ClientID and IsCurrent=1; resolve DistrictKey; update role; insert missing.                        | 5,369                   |
| DimTransactionType | Match by type code; standardise English description and cash-flow direction.                                | 4                       |
| DimOperation       | Match by operation code; standardise English description; preserve Unknown member.                          | 6                       |
| DimLoanStatus      | Match by status code; load English description, status grouping and problem flag.                           | 5                       |

Because Member 2 supplied a restored warehouse backup that already contained some records, several insert counts were correctly zero on rerun. This does not indicate a failed load; it demonstrates idempotent ETL behaviour, because rerunning the scripts updates or reconciles existing rows without creating duplicates.

## 5.5 Fact Loading

After the dimensions were available, the two fact tables were loaded using surrogate-key lookups. FactLoan was loaded using update/insert logic, while FactTransaction was treated as append-only and inserted only transactions that were not already present.

| Fact table      | Grain and key resolution                                                                                            | Measures / result                                                                                                        |
|-----------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| FactLoan        | One row per loan. Resolves LoanDateKey, AccountKey, DistrictKey and LoanStatusKey before loading.                   | LoanCount, LoanAmount, Duration, MonthlyPayment, ExpectedPaymentAmount, ProblemLoanCount. 682 rows.                      |
| FactTransaction | One row per individual transaction. Resolves DateKey, AccountKey, DistrictKey, TransactionTypeKey and OperationKey. | TransactionCount, TransactionAmount, InflowAmount, OutflowAmount, SignedAmount, BalanceAfterTransaction. 1,056,320 rows. |

The transaction fact load is implemented in 18\_load\_fact\_transaction.sql. The incremental condition prevents duplicate TransactionID values from being inserted when the ETL is rerun.

|                                      |                       |                       |                        |                            |                      |                |
|--------------------------------------|-----------------------|-----------------------|------------------------|----------------------------|----------------------|----------------|
| 100 % <span>✓ No issues found</span> |                       |                       |                        |                            |                      |                |
| Results Messages                     |                       |                       |                        |                            |                      |                |
|                                      | TotalFactTransactions | TotalTransactionCount | TotalTransactionAmount | TotalInflow                | TotalOutflow         | NetCashFlow    |
| 1                                    | 1056320               | 1056320               | 6257793560.30          | 3227472424.50              | 3030321135.80        | 197151288.70   |
| MissingTransactionsAfterLoad         |                       |                       |                        |                            |                      |                |
| 1                                    | 0                     |                       |                        |                            |                      |                |
| TransactionID DuplicateCount         |                       |                       |                        |                            |                      |                |
| MissingBalancesPreserved             |                       |                       |                        |                            |                      |                |
| 1                                    | 2999                  |                       |                        |                            |                      |                |
|                                      | InvalidDateKeys       | InvalidAccountKeys    | InvalidDistrictKeys    | InvalidTransactionTypeKeys | InvalidOperationKeys |                |
| 1                                    | 0                     | 0                     | 0                      | 0                          | 0                    |                |
|                                      | TransactionTypeName   | CashFlowDirection     | TransactionRecords     | TotalInflow                | TotalOutflow         | NetAmount      |
| 1                                    | Debit                 | Outgoing              | 634571                 | 0.00                       | 2821717377.80        | -2821717377.80 |
| 2                                    | Credit                | Incoming              | 405083                 | 3227472424.50              | 0.00                 | 3227472424.50  |
| 3                                    | Withdrawal            | Outgoing              | 16666                  | 0.00                       | 208603758.00         | -208603758.00  |

**Figure 3: FactTransaction load validation confirming 1,056, 320 records.**

100 %

No issues found

Results Messages

|   |                |                |                   |                 |
|---|----------------|----------------|-------------------|-----------------|
|   | TotalFactLoans | TotalLoanCount | TotalProblemLoans | TotalLoanAmount |
| 1 | 682            | 682            | 76                | 103261740.00    |

|   |                       |
|---|-----------------------|
|   | MissingLoansAfterLoad |
| 1 | 0                     |

|  |        |                |
|--|--------|----------------|
|  | LoanID | DuplicateCount |
|--|--------|----------------|

|   |                    |                     |                   |                 |
|---|--------------------|---------------------|-------------------|-----------------|
|   | InvalidAccountKeys | InvalidDistrictKeys | InvalidStatusKeys | InvalidDateKeys |
| 1 | 0                  | 0                   | 0                 | 0               |

|   |                |                                           |             |              |                 |
|---|----------------|-------------------------------------------|-------------|--------------|-----------------|
|   | LoanStatusCode | LoanStatusDescription                     | LoanRecords | ProblemLoans | TotalLoanAmount |
| 1 | A              | Completed Without Repayment Problems      | 203         | 0            | 18603216.00     |
| 2 | B              | Completed With Repayment Problems         | 31          | 31           | 4362348.00      |
| 3 | C              | Active Without Current Repayment Problems | 403         | 0            | 69078372.00     |
| 4 | D              | Active With Current Debt                  | 45          | 45           | 11217804.00     |

**Figure 4: FactLoan load validation confirming 682 loans.**

## 5.6 ETL Validation and Data-Quality Evidence

Validation was performed after staging, transformation and warehouse loading. The checks focused on completeness, duplicate prevention, null handling and referential integrity. The final end-to-end validation reported PASS results and zero orphan fact relationships.

| Validation item                                              | Result                              |
|--------------------------------------------------------------|-------------------------------------|
| Transactions preserved from clean staging to FactTransaction | 1,056,320                           |
| Loans preserved from clean staging to FactLoan               | 682                                 |
| Problem loans retained for analysis                          | 76                                  |
| Duplicate TransactionID / LoanID records                     | 0                                   |
| Invalid fact-to-dimension relationships                      | 0                                   |
| Missing balances                                             | 2,999 preserved as NULL             |
| Missing transaction operations                               | 183,114 mapped to Unknown Operation |
| Final ETL validation status                                  | PASS                                |

These checks show that the ETL process preserved the operational records while applying the transformations required for analytical reporting. No business records were intentionally discarded simply because a value was unknown or represented an adverse loan outcome.

| Results Messages |                                            |                |               |                  |
|------------------|--------------------------------------------|----------------|---------------|------------------|
|                  | ValidationCheck                            | SourceRows     | TargetRows    | ValidationStatus |
| 1                | Account transformation                     | 4500           | 4500          | PASS             |
| 2                | Client transformation                      | 5369           | 5369          | PASS             |
| 3                | District transformation                    | 77             | 77            | PASS             |
| 4                | Loan transformation                        | 682            | 682           | PASS             |
| 5                | Transaction transformation                 | 1056320        | 1056320       | PASS             |
| 6                | Loan fact loading                          | 682            | 682           | PASS             |
| 7                | Transaction fact loading                   | 1056320        | 1056320       | PASS             |
|                  | ValidationCheck                            | InvalidRows    |               |                  |
| 1                | FactLoan invalid AccountKey                | 0              |               |                  |
| 2                | FactLoan invalid DistrictKey               | 0              |               |                  |
| 3                | FactLoan invalid LoanStatusKey             | 0              |               |                  |
| 4                | FactTransaction invalid AccountKey         | 0              |               |                  |
| 5                | FactTransaction invalid DistrictKey        | 0              |               |                  |
| 6                | FactTransaction invalid TransactionTypeKey | 0              |               |                  |
| 7                | FactTransaction invalid OperationKey       | 0              |               |                  |
|                  | ValidationCheck                            | WarehouseValue | DataMartValue | ValidationStatus |
| 1                | Transaction data mart                      | 1056320        | 1056320       | PASS             |
| 2                | Loan data mart                             | 682            | 682           | PASS             |
| 3                | Problem-loan data ...                      | 76             | 76            | PASS             |

**figure 5: final ETL validation showing row counts, duplicates, and referential integrity checks all passed.**

## 5.7 ETL Script Execution Order

The ETL scripts are executed in numeric order so dependencies are clear and reproducible. The sequence first verifies the restored warehouse, then creates and profiles staging, transforms the five warehouse subjects, loads dimensions, loads facts, creates the data mart and finally runs end-to-end validation.

| Sequence | Purpose                                                                                   |
|----------|-------------------------------------------------------------------------------------------|
| 01–03    | Verify restored warehouse and inspect source / warehouse structures.                      |
| 04–05    | Create raw staging tables and profile the staged data.                                    |
| 06–11    | Transform account, district, client, loan and transaction subjects.                       |
| 12–16    | Inspect load structure and load DimDistrict, DimAccount, DimClient and lookup dimensions. |
| 17–18    | Load FactLoan and FactTransaction.                                                        |
| 19       | Create and reconcile the Retail Banking Performance Data Mart.                            |
| 20       | Run final end-to-end ETL validation.                                                      |

## Member 3 Deliverable: Task 6 – Data Mart Development and Reconciliation

### 6.1 Data Mart Purpose, Target Users and Benefits

A subject-focused Retail Banking Performance Data Mart was created under the mart schema as a reporting layer derived from the enterprise warehouse. Its main purpose is to simplify Power BI analysis, centralise recurring business calculations and avoid repeatedly aggregating more than one million detailed transaction rows.

- Target users: branch/regional managers, lending or risk analysts, finance analysts and BI/reporting users.
- Analytical benefit: faster access to prepared monthly transaction and district-level loan summaries.
- Consistency benefit: the same aggregation logic is reused across dashboards instead of being recreated independently in Power BI.
- Usability benefit: business-facing measures are available in a simpler structure than the detailed warehouse facts.

### 6.2 Data Mart Implementation

The data-mart implementation is contained in 19\_create\_retail\_banking\_data\_mart.sql and creates two reporting tables:

| Data-mart table                    | Grain                      | Key measures                                                                                                | Recommended analysis                                          |
|------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| mart.MonthlyTransactionPerformance | One row per year and month | Transactions, active accounts, total amount, inflow, outflow, net cash flow and average transaction amount. | Monthly trends, KPI cards and cash-flow monitoring.           |
| mart.DistrictLoanPerformance       | One row per district       | Loans, borrowing accounts, total/average loan amount, payments, problem loans and problem-loan percentage.  | Regional comparison, exposure ranking and loan-risk analysis. |

### 6.3 MonthlyTransactionPerformance

mart.MonthlyTransactionPerformance aggregates the detailed FactTransaction data to one row per calendar month. It supports time-series reporting without requiring every dashboard visual to scan the full transaction fact table.

- Year, quarter, month and year-month label for trend analysis.
- Total transaction count and number of active accounts.
- Total transaction amount and average transaction amount.
- Total inflow, total outflow and net cash flow.

### 6.4 DistrictLoanPerformance

mart.DistrictLoanPerformance aggregates FactLoan by district and combines loan measures with regional attributes. This supports comparison of loan exposure and repayment risk across districts and can be used together with demographic or economic measures.

- District and region identifiers.
- Total loans and borrowing accounts.
- Total and average loan amount.
- Monthly and expected payment amounts.
- Problem-loan count and problem-loan percentage.
- Regional population and economic indicators for comparative analysis.

## 6.5 Refresh Strategy and Reconciliation

The data-mart script uses a controlled full-refresh approach for the two derived mart tables: it drops and recreates only the mart summaries while leaving the enterprise warehouse dimensions and facts unchanged. This makes the summaries reproducible whenever the warehouse data changes.

Reconciliation was then used to compare the aggregated mart totals with the underlying warehouse totals. Matching figures confirm that the aggregation process did not lose or duplicate records.

| Comparison    | Warehouse total | Data-mart total | Status |
|---------------|-----------------|-----------------|--------|
| Transactions  | 1,056,320       | 1,056,320       | PASS   |
| Loans         | 682             | 682             | PASS   |
| Problem loans | 76              | 76              | PASS   |

| DataMartTable                   | SummaryRows | ReconciledRecordCount |
|---------------------------------|-------------|-----------------------|
| 1 MonthlyTransactionPerformance | 72          | 1056320               |
| 2 DistrictLoanPerformance       | 77          | 682                   |

| WarehouseTransactions | DataMartTransactions |
|-----------------------|----------------------|
| 1 1056320             | 1056320              |

| WarehouseLoans | DataMartLoans | WarehouseProblemLoans | DataMartProblemLoans |
|----------------|---------------|-----------------------|----------------------|
| 1 682          | 682           | 76                    | 76                   |

**Figure 6: Data mart reconciliation confirming that warehouse and data-mart totals match for transactions, loans, and problem loans.**

## 6.6 Member 3 Final Outputs and Handover

The completed Member 3 work provides Member 4 with a validated warehouse and ready-to-use reporting layer. Power BI can therefore focus on semantic modelling, measures, interactive visuals, findings and recommendations instead of repeating the ETL logic.

- Complete raw and clean staging workflow for the warehouse subjects.
- Repeatable dimension and fact loading with surrogate-key resolution and duplicate-safe behaviour.
- Validated FactTransaction with 1,056,320 transactions and FactLoan with 682 loans.
- Retail Banking Performance Data Mart with monthly transaction and district loan summaries.
- Reconciliation evidence confirming 1,056,320 transactions, 682 loans and 76 problem loans in both warehouse and mart.
- Final end-to-end ETL validation showing PASS results and no invalid fact-to-dimension relationships.

## 6.7 Summary of Member 3 Contribution

In summary, Member 3 implemented the end-to-end data movement from staged source files into analytical warehouse structures. The work included extraction and staging, profiling, cleansing and standardisation, derived business measures, dimension and fact loading, data-quality validation, data-mart creation and reconciliation. The resulting warehouse and Retail Banking Performance Data Mart provide a consistent and validated foundation for the Power BI analysis completed in the next project stage.

# Power BI Evidence

## Task 7 overview: connection, model and dashboard

The Power BI Desktop was then connected to the SQL Server database CzechBank\_DW (localhost) via Import mode. The fact and dimension tables were created by Member 2 and loaded by Member 3 through an ETL process, with mart.MonthlyTransactionPerformance and mart.DistrictLoanPerformance being the two additional pre-aggregated data-mart tables. The dashboard pages utilize mart tables which ensure that the visuals remain quick with more than 1,056,320 transaction rows whereas the fact loan and its dimensions are used for loan analysis.

Eight DAX measures were then created (Table 1) and there are three pages in the report: Executive Summary, Trend Analysis and Interactive Analysis. Drill down and roll up features have been enabled for the Region/District visualization. The totals have been reconciled against the warehouse via SQL (Figures 17 and 18).

### 1. Connection / Navigator

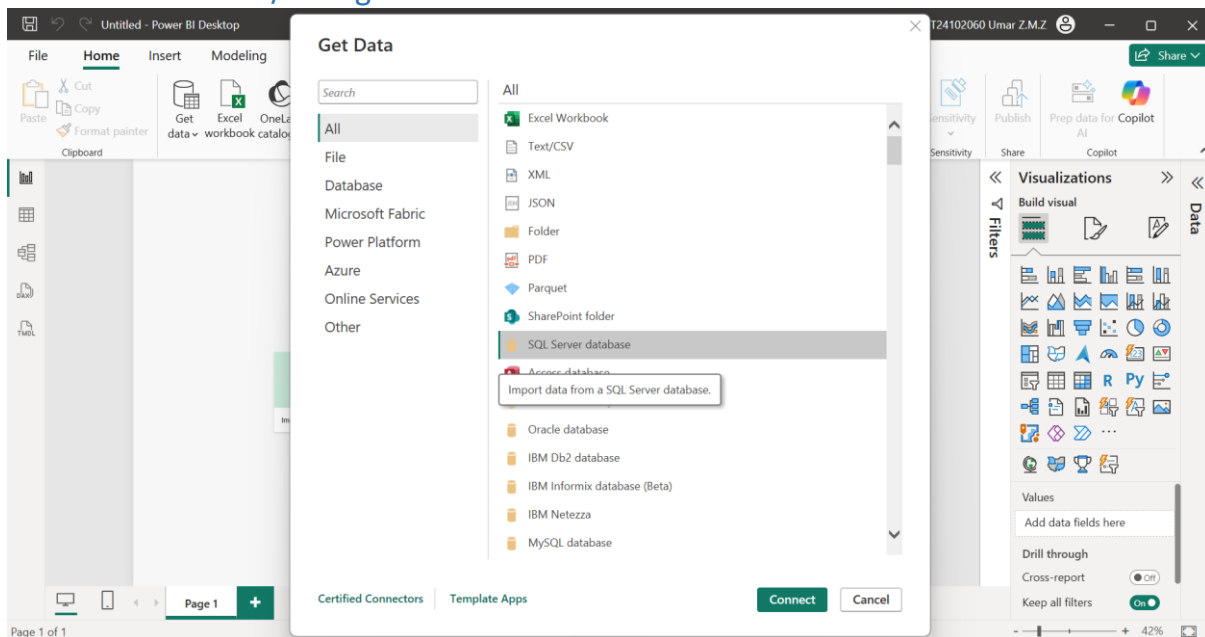


Figure 1: Get Data in Power BI Desktop with the SQL Server database connector selected



Figure 2: Connection to server localhost, database CzechBank\_DW, using Import mode

## Navigator

Display Options ▾

☐ card
 ☐ client
 ☒ DimAccount
 ☒ DimClient
 ☒ DimDate
 ☒ DimDistrict
 ☒ DimLoanStatus
 ☒ DimOperation
 ☒ DimTransactionType
 ☐ disp
 ☐ district
 ☒ FactLoan
 ☒ FactTransaction
 ☐ loan
 ☒ mart.DistrictLoanPerformance
 ☒ mart.MonthlyTransactionPerformance
 ☐ order
 ☐ stg.account
 ☐ stg.account clean

mart.MonthlyTransactionPerformance

| Year | MonthNumber | MonthName | Quarter | YearMonthLabel | TotalTransa |
|------|-------------|-----------|---------|----------------|-------------|
| 1993 | 1           | January   | 1       | 1993-01        |             |
| 1993 | 2           | February  | 1       | 1993-02        |             |
| 1993 | 3           | March     | 1       | 1993-03        |             |
| 1993 | 4           | April     | 2       | 1993-04        |             |
| 1993 | 5           | May       | 2       | 1993-05        |             |
| 1993 | 6           | June      | 2       | 1993-06        |             |
| 1993 | 7           | July      | 3       | 1993-07        |             |
| 1993 | 8           | August    | 3       | 1993-08        |             |
| 1993 | 9           | September | 3       | 1993-09        |             |
| 1993 | 10          | October   | 4       | 1993-10        |             |
| 1993 | 11          | November  | 4       | 1993-11        |             |
| 1993 | 12          | December  | 4       | 1993-12        |             |
| 1994 | 1           | January   | 1       | 1994-01        |             |
| 1994 | 2           | February  | 1       | 1994-02        |             |
| 1994 | 3           | March     | 1       | 1994-03        |             |
| 1994 | 4           | April     | 2       | 1994-04        |             |
| 1994 | 5           | May       | 2       | 1994-05        |             |
| 1994 | 6           | June      | 2       | 1994-06        |             |
| 1994 | 7           | July      | 3       | 1994-07        |             |
| 1994 | 8           | August    | 3       | 1994-08        |             |
| 1994 | 9           | September | 3       | 1994-09        |             |
| 1994 | 10          | October   | 4       | 1994-10        |             |

Select Related Tables

Load

Transform Data

Cancel

Figure 3: Navigator showing the dimension, fact and data-mart tables selected for loading.

Data >>

Search

>

 DimAccount
 

>

 DimClient
 

>

 DimDate
 

>

 DimDistrict
 

>

 DimLoanStatus
 

>

 DimOperation
 

>

 DimTransactionType
 

>

 FactLoan
 

>

 FactTransaction
 

>

 mart DistrictLoanPerfo...
 

>

 mart MonthlyTransacti...

Figure 4: Data pane listing the tables loaded into the model

## 2. Model View

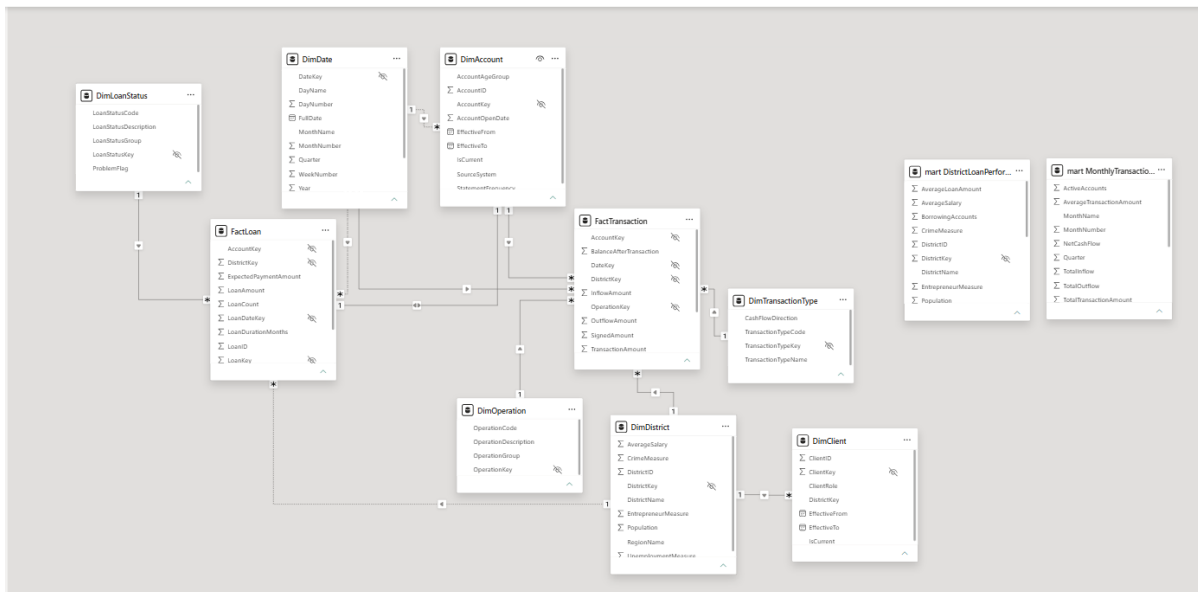


Figure 5

Model view. Fact tables connect to dimension tables through surrogate keys (mostly many-to-one with single-direction filtering); the two mart tables are standalone reporting tables

### 3. Measures List And Hierarchies for OLAP

#### mart MonthlyTransactionPerformance

|                        |                                                                           |
|------------------------|---------------------------------------------------------------------------|
| Total Transactions     |                                                                           |
| 1 Total Transactions = | <code>SUM('mart MonthlyTransactionPerformance'[TotalTransactions])</code> |
| Total Inflow           |                                                                           |
| 1 Total Inflow =       | <code>SUM('mart MonthlyTransactionPerformance'[TotalInflow])</code>       |
| Total Outflow          |                                                                           |
| 1 Total Outflow =      | <code>SUM('mart MonthlyTransactionPerformance'[TotalOutflow])</code>      |
| Net Cash Flow          |                                                                           |
| 1 Net Cash Flow =      | <code>SUM('mart MonthlyTransactionPerformance'[NetCashFlow])</code>       |

Figure 6: Measures defined in mart MonthlyTransactionPerformance

#### mart DistrictLoanPerformance

|                       |                                                                                  |
|-----------------------|----------------------------------------------------------------------------------|
| Total Loans           |                                                                                  |
| 1 Total Loans =       | <code>SUM('mart DistrictLoanPerformance'[TotalLoans])</code>                     |
| Total Loan Amount     |                                                                                  |
| 1 Total Loan Amount = | <code>SUM('mart DistrictLoanPerformance'[TotalLoanAmount])</code>                |
| Problem Loans         | 1 Problem Loans = <code>SUM('mart DistrictLoanPerformance'[ProblemLoans])</code> |
| Problem Loan Rate     | 1 Problem Loan Rate = <code>DIVIDE([Problem Loans], [Total Loans], 0)</code>     |

Figure 7: Measures defined in mart DistrictLoanPerformance

Changed the format to percentage for Problem Loan Rate Measure

TABLE: Measure Definitions

| MEASURE | DAX FORMULA | Purpose and Where Used |
|---------|-------------|------------------------|
|---------|-------------|------------------------|

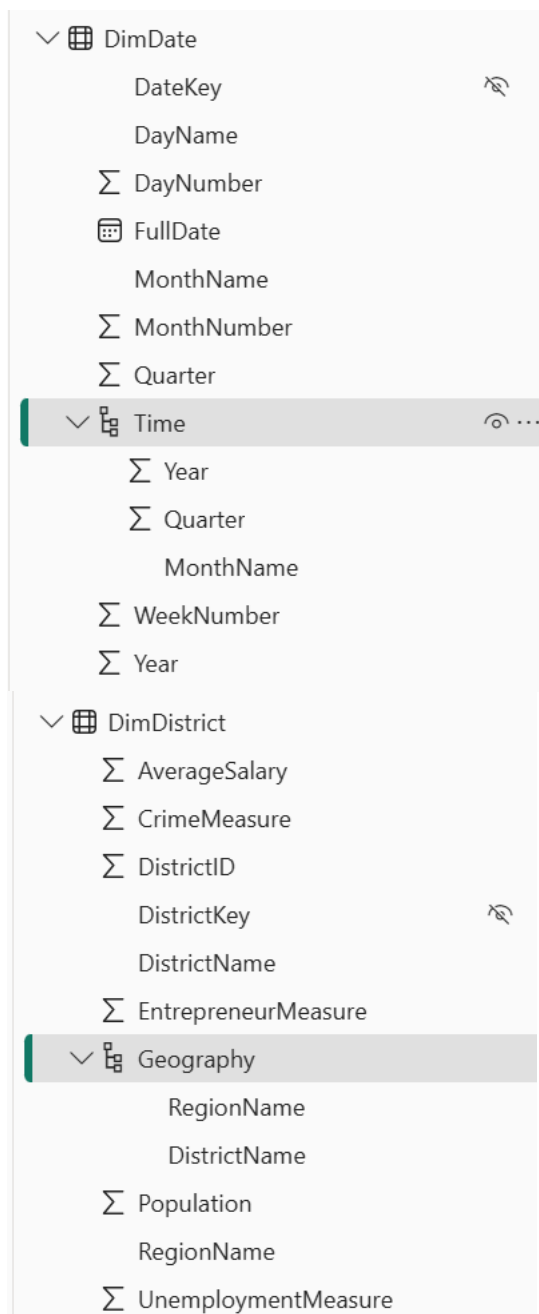
|                    |                                                               |                                                                                           |
|--------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Total Transactions | SUM('mart MonthlyTransactionPerformance'[TotalTransactions])) | Overall transaction volume. Page 1 card, Page 2 line chart.                               |
| Total Inflow       | SUM('mart MonthlyTransactionPerformance'[TotalInflow]))       | Money received into accounts. Page 1 card, Page 2 columns.                                |
| Total Outflow      | SUM('mart MonthlyTransactionPerformance'[TotalOutflow]))      | Money paid out of accounts. Page 1 card, Page 2 columns.                                  |
| Net Cash Flow      | SUM('mart MonthlyTransactionPerformance'[NetCashFlow]))       | Inflow minus outflow; liquidity indicator. Page 1 card.                                   |
| Total Loans        | SUM('mart DistrictLoanPerformance'[TotalLoans]))              | Number of loans issued. Page 1 card, Page 3 table.                                        |
| Total Loan Amount  | SUM('mart DistrictLoanPerformance'[TotalLoanAmount]))         | Value of loans issued (exposure). Page 3 bar chart.                                       |
| Problem Loans      | SUM('mart DistrictLoanPerformance'[ProblemLoans]))            | Loans with repayment problems (status B and D). Page 1 card, Page 3 table.                |
| Problem Loan Rate  | DIVIDE([Problem Loans], [Total Loans], 0)                     | Share of loans in a problem status; formatted as a percentage. Page 1 card, Page 3 table. |

## Hierarchies for OLAP

Hierarchies allow users to drill down and roll up through levels of detail. The following hierarchies support the OLAP analysis on Pages 2 and 3 (Table 2). The Region / District hierarchy is used for drill-down and roll-up in Figures 13 to 16.

**TABLE:** Hierarchies defined in the model.

| Hierarchy | Table       | Levels                     |
|-----------|-------------|----------------------------|
| Time      | DimDate     | Year → Quarter → MonthName |
| Geography | DimDistrict | RegionName → DistrictName  |

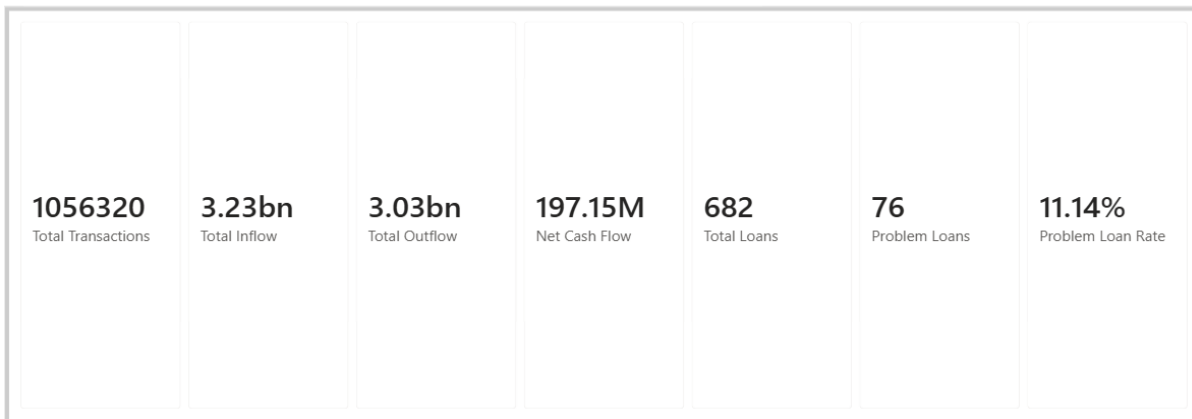


**Figure 8:** Evidence of Hierarchy

#### 4. Three Pages

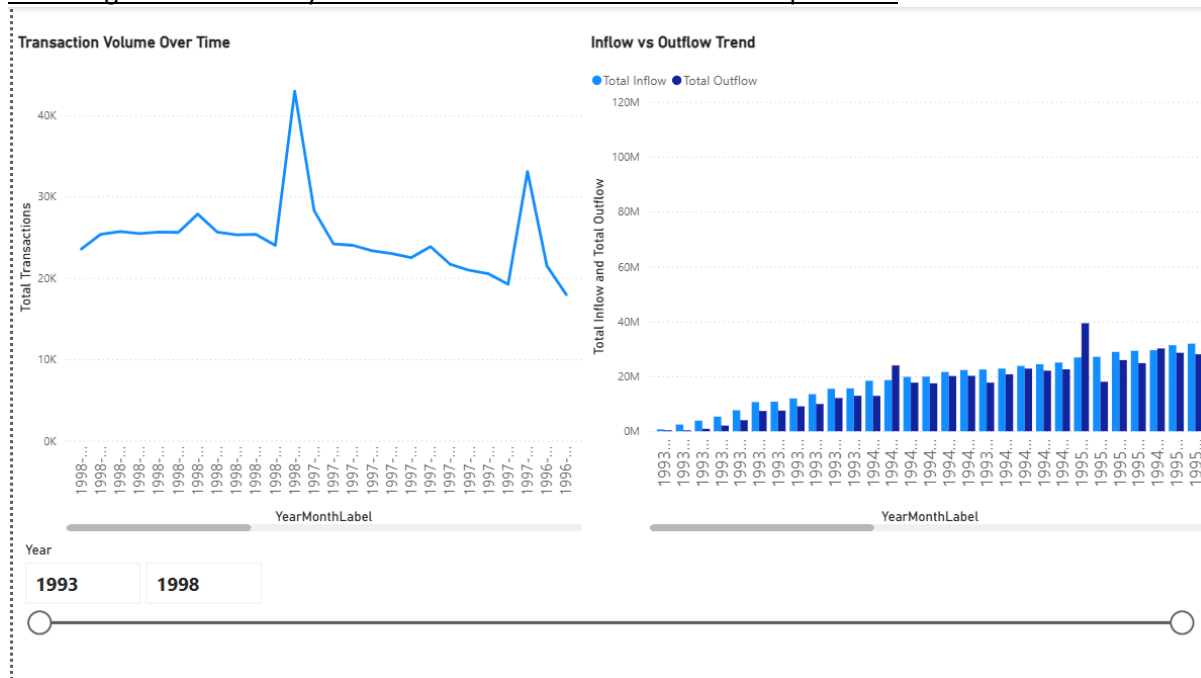
Build Page 1 - Executive Summary with KPI cards and overall performance indicators

## Retail Banking Performance – Executive Summary



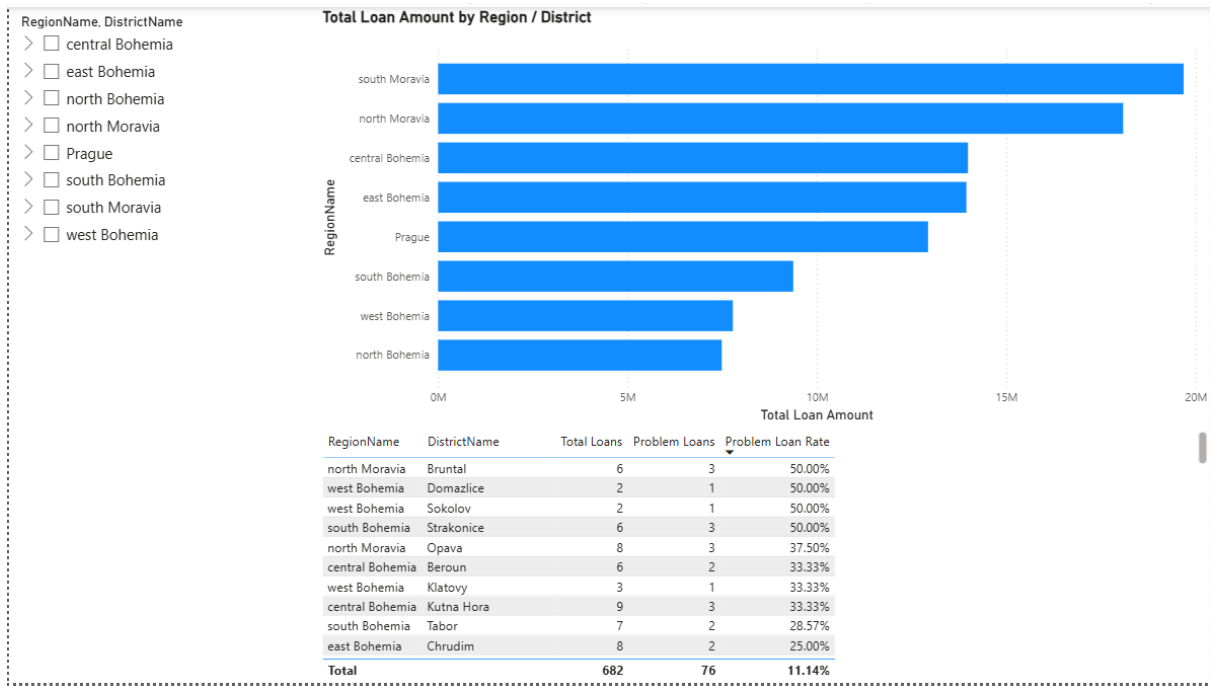
**Figure 9:** Page 1 – Executive Summary: KPI cards for Total Transactions (1,056,320), Total Inflow (3.23bn), Total Outflow (3.03bn), Net Cash Flow (197.15M), Total Loans (682), Problem Loans (76) and Problem Loan Rate (11.14%).

### Build Page 2 - Trend Analysis with time-based line charts and comparisons.



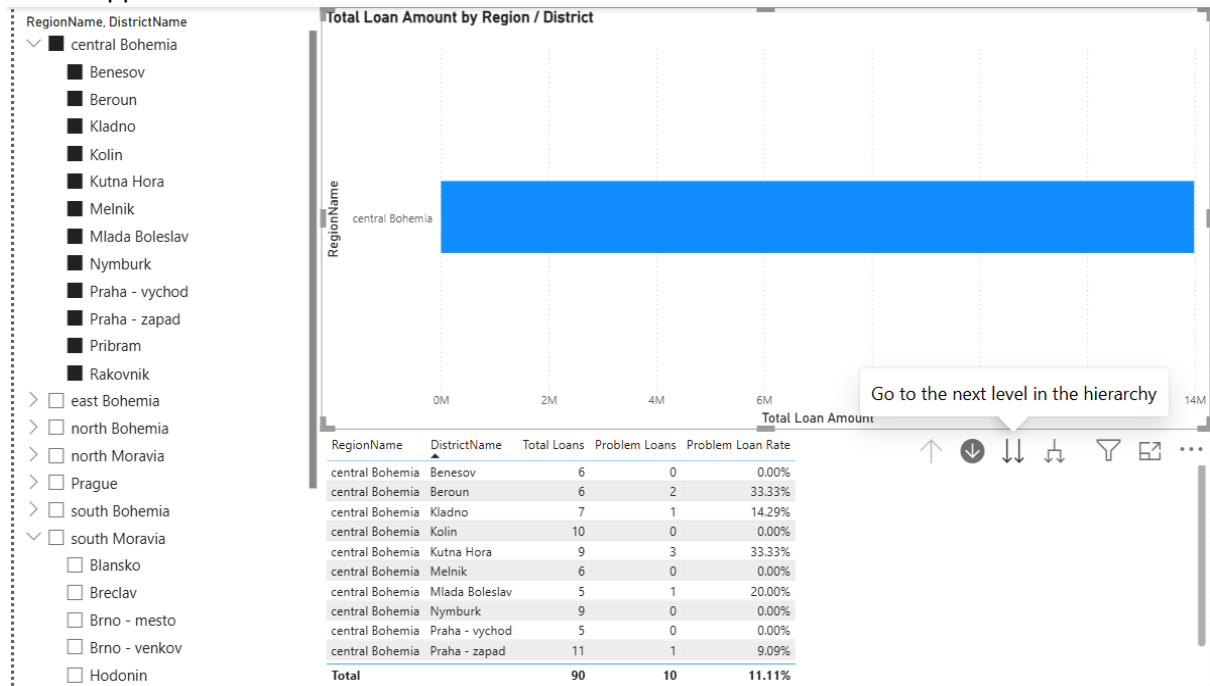
**Figure 10:** Page 2 – Trend Analysis: monthly transaction volume (line chart) and inflow vs outflow (clustered columns), with a Year slicer

### Build Page 3 - Interactive Analysis with filters, slicers, drill-down and roll-up.



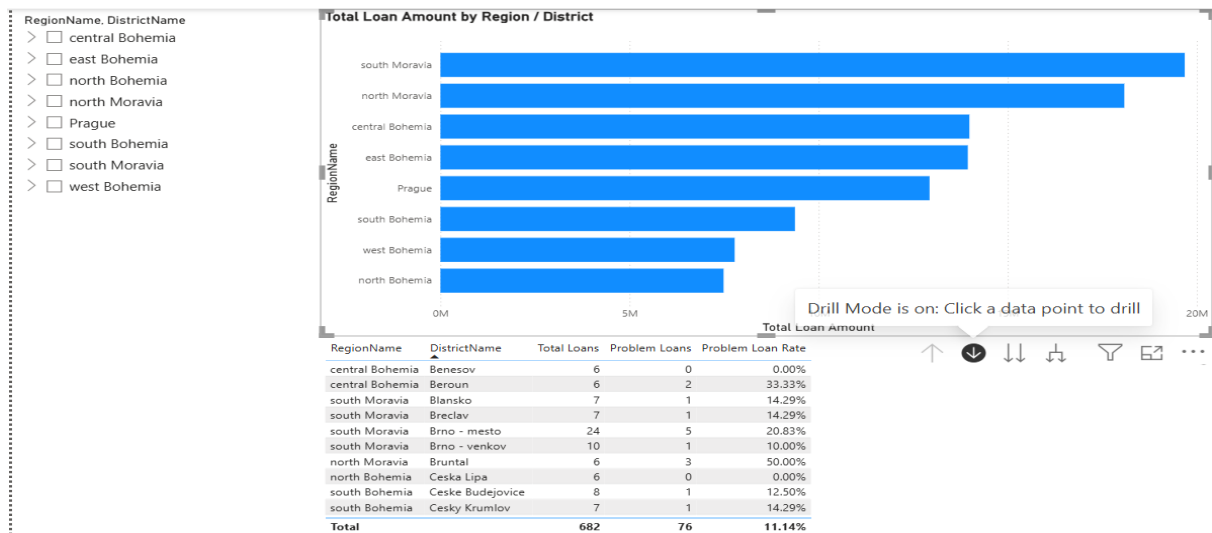
**Figure 11:** Page 3 – Interactive Analysis: Region / District slicer, drill-down bar chart of Total Loan Amount, and detail table of loans, problem loans and problem loan rate.

## Slicers Applied

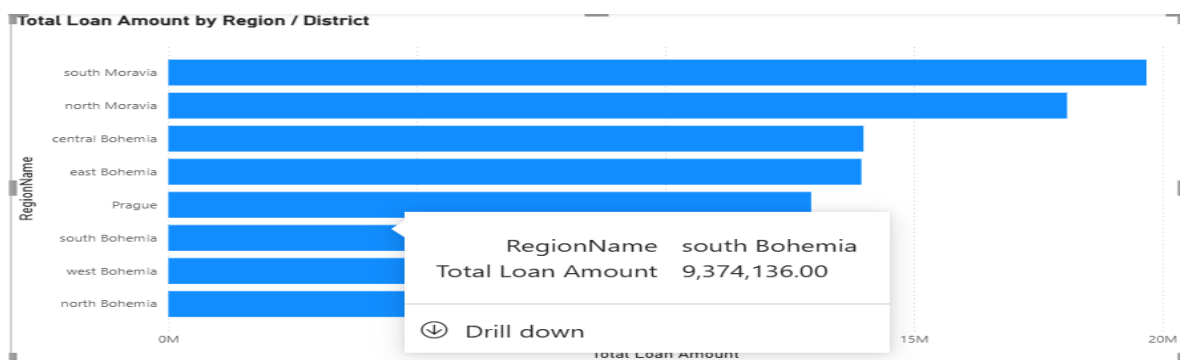


**Figure 12:** Slicer applied: central Bohemia selected, so the chart and table filter to that region (90 loans, 10 problem loans, 11.11%).

## Drill-mode

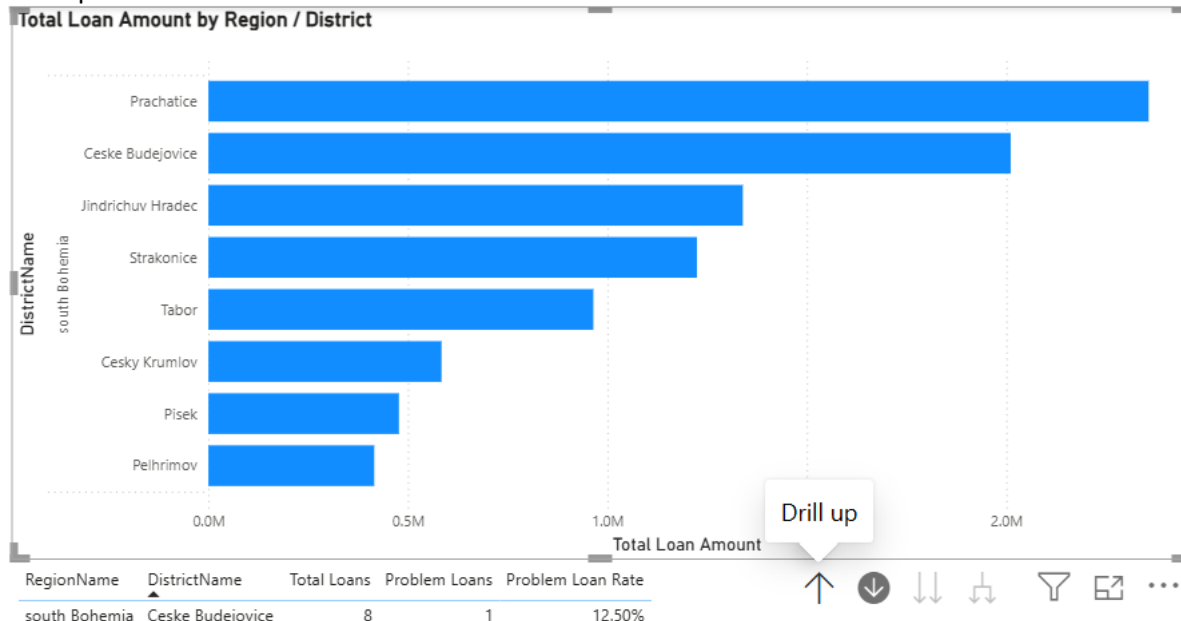


**Figure 13:** Drill mode switched on in the chart header.

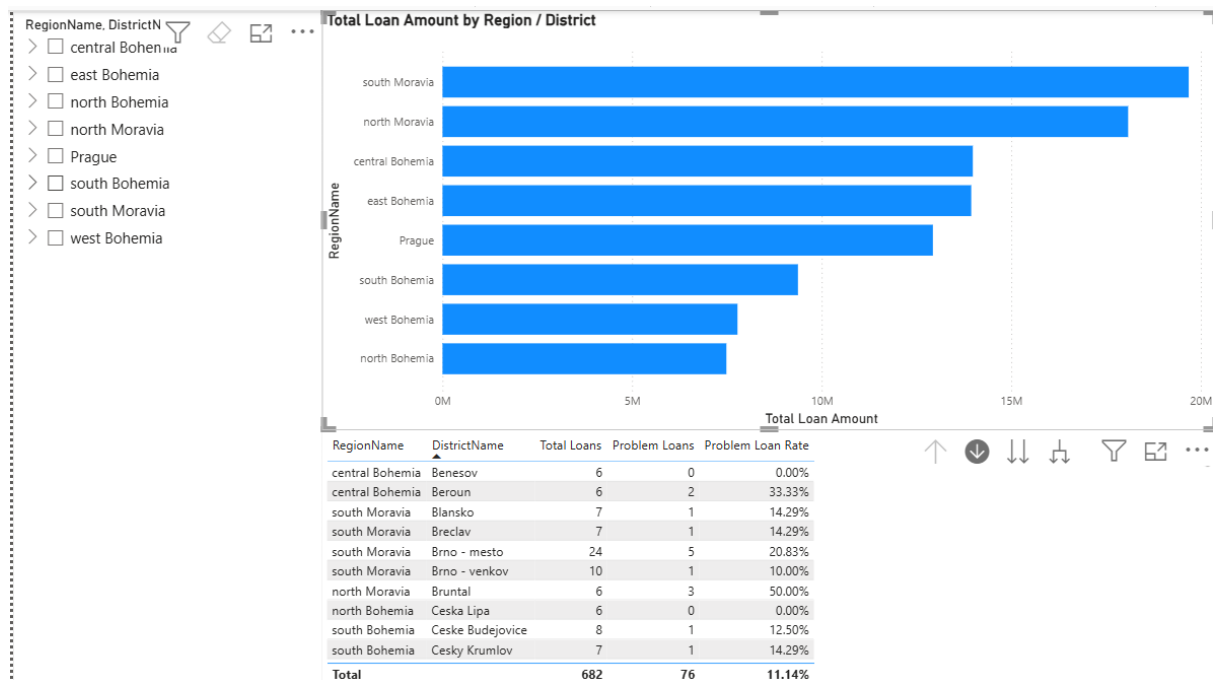


**Figure 14:** Drill down: selecting the south Bohemia bar and choosing Drill down.

## Roll-up



**Figure 15:** Result of the drill-down: districts within south Bohemia, with the Drill up button available.

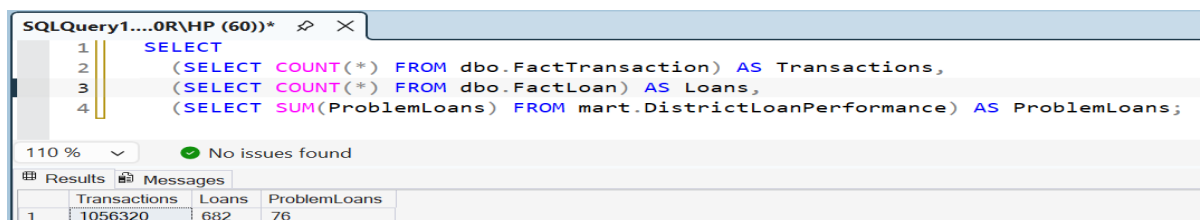


**Figure 16:** Roll-up: after Drill up the chart returns to region level.

## 5. Power BI totals match SQL warehouse totals

| Total Transactions | Total Loans | Problem Loans |
|--------------------|-------------|---------------|
| 1056320            | 682         | 76            |

**Figure 17:** Power BI table showing 1,056,320 transactions, 682 loans and 76 problem loans.



```

1 SELECT
2 (SELECT COUNT(*) FROM dbo.FactTransaction) AS Transactions,
3 (SELECT COUNT(*) FROM dbo.FactLoan) AS Loans,
4 (SELECT SUM(ProblemLoans) FROM mart.DistrictLoanPerformance) AS ProblemLoans;

```

|   | Transactions | Loans | ProblemLoans |
|---|--------------|-------|--------------|
| 1 | 1056320      | 682   | 76           |

Figure 18: SQL query on CzechBank\_DW returning the same totals.

Both tools return the same values, so the dashboard totals reconcile to the warehouse and data mart

## Insights and Recommendations

Seven insights were derived from the dashboard pages

### 1. The bank is cash-positive overall, but the margin is thin

Evidence (Page 1): Inflow is 3.23bn while outflow is 3.03bn, resulting in net cash flow of 197.15M, which is just around 6% of inflows.

Recommendation: Monitor net cash flow on a monthly basis as one of key liquidity indicators. Establish internal minimum level of net cash flow as a percentage of inflows.

### 2. The transactions have been increasing at an extremely rapid rate during this period.

Evidence (Page 2 line chart): Transactions increase from 177 per month in January 1993 to around 25,000 per month by the end of 1998. Total transactions equal 1,056,320 (Page 1).

Recommendation: Plan for capacity, support staff, and fraud control based on continuing volume increases. Utilize increased customer base to cross-sell.

### 3. January is the most active month.

Evidence (Page 2 line graph): The highest peaks are found in January. Jan 1998 has a value of 42,940 transactions, while Jan 1997 has a value of 33,060.

Recommendation: Staff and system should be expanded before January comes around, and maintenance can be done during February – April..

### 4. The cash flow deficits occur only in January and June, but those in January are increasing.

Evidence (page 2 inflows vs outflows columns): There are 10 instances where outflow is higher than inflow in 72 months, but all 10 instances occur in either January or June. The January deficit gap keeps widening every year, from an estimated -5.4M in 1994 to an estimated -33.5M in 1998.

Recommendation: Prepare a liquidity cushion before every January and June based on the last year's deficit gap (at least 35M in January 1999). Find the cause of the huge January outflow.

### 5. There are only a few locations with loan exposure.

Evidences (Page 3 bar chart): Hl.m. Praha has the highest exposure of around 12.9M, 3 times greater than the second highest district (Brno - mesto at around 4.0M). On a region basis, south Moravia (around 19.7M) and north Moravia (around 18.1M) have the highest exposures.

Recommendation: Establish maximum exposure thresholds on district and regional basis and monitor on a monthly basis. Start with south and north Moravia portfolios first.

### 6. The problem loan risks show wide regional variations, and small districts may give misleading results.

Evidence (Page 3 table and slicer): The average figure stands at 11.14% (76 out of 682 loans). Choose any region using the slicer, and the total in the table changes accordingly. The figure for north Moravia is 15.4% (18 out of 117), while north Bohemia has only 1.6% (1 out of 61). From the table, Bruntal comes in at 50% but on just six loans.

Recommendation: Rank or flag districts only if they have ten or more loans. Credit check priority goes to north Moravia because it has the highest problem loans in the regions.

## 7. A few high-exposure areas require a watchlist, whereas Prague is relatively secure.

Evidence (Table on page 3, ordered according to Problem Loan Percentage): In Brno - mesto, 24 loans include 5 problem loans, representing 20.83% of total loans, which is almost twice the bank average. Svitavy has 3 problem loans among 12 total loans (25%) and Ostrava - mesto has 3 out of 20 (15%). Prague, despite the highest exposure, has only 7 out of 84 (8.33%) problem loans, below the bank average.

Recommendation: Create a watchlist for Brno - mesto, Svitavy and Ostrava - mesto with a stricter loan approval process. Consider Prague as a large but relatively low-risk exposure.

### Demo plan

**TABLE: Four-Minute Demo Flow**

| Time      | Page     | What to show                                                                             | Key message                                                                                     |
|-----------|----------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 0:00–0:30 | Page 1   | KPI cards; Figures 17–18 reconciliation                                                  | Bank is cash-positive (197.15M net) and dashboard totals match the warehouse.                   |
| 0:30–1:30 | Page 2   | Line chart and inflow vs outflow (full 1993–1998 range)                                  | Volume grows strongly, January peaks, deficits occur only in January and June and are widening. |
| 1:30–3:00 | Page 3   | Live: pick a region in the slicer, drill down to districts, drill up, sort table by rate | Exposure is concentrated and problem-loan risk differs sharply by region.                       |
| 3:00–4:00 | Insights | Insights 4, 5 and 7                                                                      | Recommendations: liquidity buffer, exposure limits and a credit watchlist.                      |

### Acknowledgements

The dimensional warehouse (dimensions, fact tables, keys, relationships and database backup) was designed and built by Member 2. The ETL process, data quality validation and the Retail Banking Performance Data Mart were implemented by Member 3. This section (Tasks 7 and 8) builds on those outputs and does not repeat their transformations.

### AI-use acknowledgement

AI tools were used as a supporting tool for learning and development: to explain Power BI features, suggest page layouts and DAX patterns, and help with the insights format. All data connections, measures, dashboards, screenshots and final wording were produced by me, and every figure quoted in the insights was checked in Power BI and reconciled to the SQL warehouse totals before inclusion.